BOE

TITLE ：NE140QDM-NX5 V18.3

Product Specification

Rev. P0

Nanjing BOE Display Technology Co.,Ltd

SPEC. NUMBER	PRODUCT GROUP	Rev. P0	ISSUE DATE	PAGE
	TFT-LCD		2024.02.19	1 OF 70

REVISION HISTORY

(√)Preliminary Specification

( )Final Specification

Revision No.	Page	Description of Changes	Date	Prepared
P0	70	Initial Release	2024.02.19	Zhao yang

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Contents

1.0 GENERAL DESCRIPTION

1.1 Introduction

NE140QDM-NX5 is a color active matrix TFT LCD module using IGZO TFT's (Thin Film Transistors) as an active switching devices. This module has a 14.0 inch diagonally measured active area with WQX GA resolutions (2560 horizontal by 1600 vertical pixel array). Each pixel is divided into RED, GREEN, BLUE dots which are arranged in vertical stripe and this module can display 1.07B (8bit+FRC) colors a nd color gamut DCI-P3 100% typ., DCI-P3 95% min. The TFT-LCD panel used for this module is a lo w reflection and higher color type. Therefore, this module is suitable for Notebook CC. The LED driver for back-light driving is built in this model.

All input signals are eDP1.4b interface compatible.

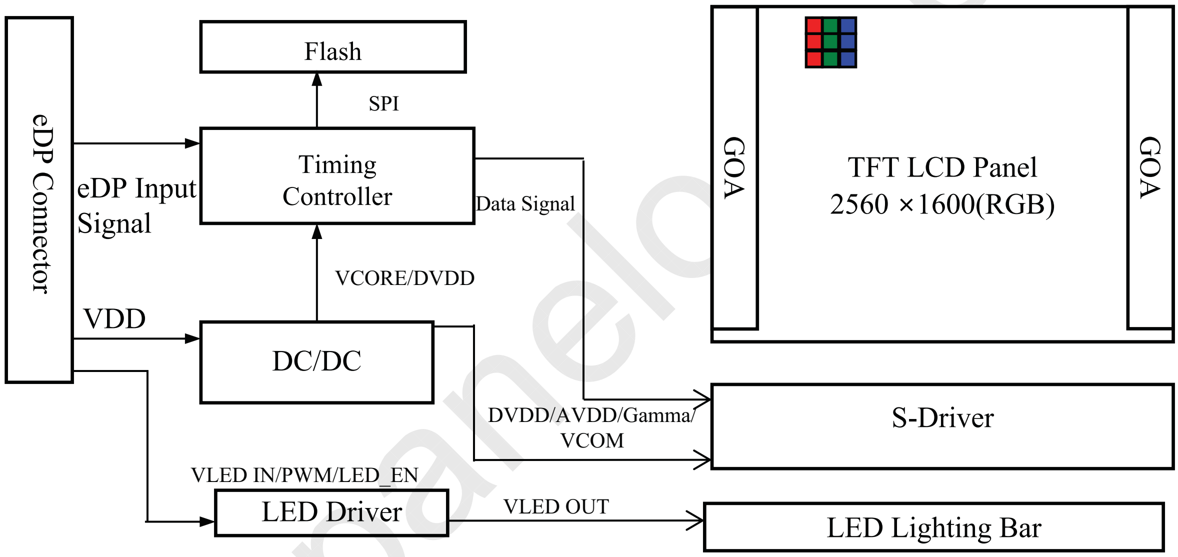


Figure 1. Drive Architecture

1.2 Features

- 4 lane eDP interface with 5.4Gbps link rates
- Thin and light weight
- 1.07B(8bit+FRC) color depth, color gamut DCI-P3 100%
- Single LED lighting bar (Bottom side/Horizontal Direction)
- Data enable signal mode
- Side mounting frame
- Green product (RoHS & Halogen free product)
- On board LED driving circuit
- Low driving voltage and low power consumption
- On board EDID chip
- Adjust backlight brightness with DC mode

1.0 GENERAL DESCRIPTION

1.2 Features

DPCD Ver.	sDRRS	DCR	DMRRS	PSR	LRR	MBO
1.4	on	off	off	PSR2	LRR2.5	off
VESA DSC	MSO	AMD Free-sync	HDR	Dimming	OD	BIST
off	off	on	HDR400	PWM in DC out	off	yes
FRC	DDS	G-sync	DPST			
on	off	yes	yes			

1.3 Application

- Notebook PC (Wide type)

1.4 General Specification

The followings are general specifications at the model NE140QDM-NX5. (listed in Table 1)

<Table 1. General Specifications>

Parameter	Specification	Unit	Remarks
Active area	301.59(H) ×188.50(V)	mm	
Number of pixels	2560 (H) ×1600 (V)	pixels	
Pixel pitch	117.81(H) ×117.81(V)	um	
Pixel arrangement	RGB Vertical stripe		
Display colors	1.07B(8bit+FRC)		
Color gamut	DCI-P3 100% typ., DCI-P3 95% min		
Display mode	Normally black		
Dimensional outline	306.59±0.3 (H)*198.4±0.5(V)*1.95(Max)(W/O PCB) 306.59±0.3(H)*198.4±0.5(V) *3.95(Max)(W/PCB)	mm	
Weight	180(max)	g	
Surface treatment	Anti-Glare		
Surface hardness	3H		
Back-light	Bottom edge side, 1-LED lighting bar type		Note 1
Power consumption	P_D : 1.1(Max.)	W	@Mosaic
	P_{BL} : 4.82(Max.)	W	@VLED=12V
	P_{Total} : 5.92(Max.)	W	@Mosaic

Notes : 1. LED Lighting Bar (60*LED Array)

2.0 ABSOLUTE MAXIMUM RATINGS

The followings are maximum values which, if exceed, may cause faulty operation or damage to the unit. The operational and non-operational maximum voltage and current values are listed in Table 2.

< Table 2. Absolute Maximum Ratings>

Ta=25+/-2°C

Parameter	Symbol	Min.	Max.	Unit	Remarks
Power Supply Voltage	V_{DD}	-0.3	4.0	V	Note 1
eDP input Voltage	V_{eDP}	0	2.0	V	
Logic Supply Voltage	V_{IN}	V_{SS}-0.3	V_{DD}+0.3	V	
Operating Temperature	T_{OP}	0	+50	°C	Note 2
Storage Temperature	T_{ST}	-20	+60	°C	

Notes :

1. Permanent damage to the device may occur if maximum values are exceeded functional operation should be restricted to the condition described under normal operating conditions.
2. Temperature and relative humidity range are shown in the figure below.

90 % RH Max. (40 °C ≥ Ta) Maximum wet - bulb temperature at 39 °C or less. (Ta > 40 °C) No condensation.

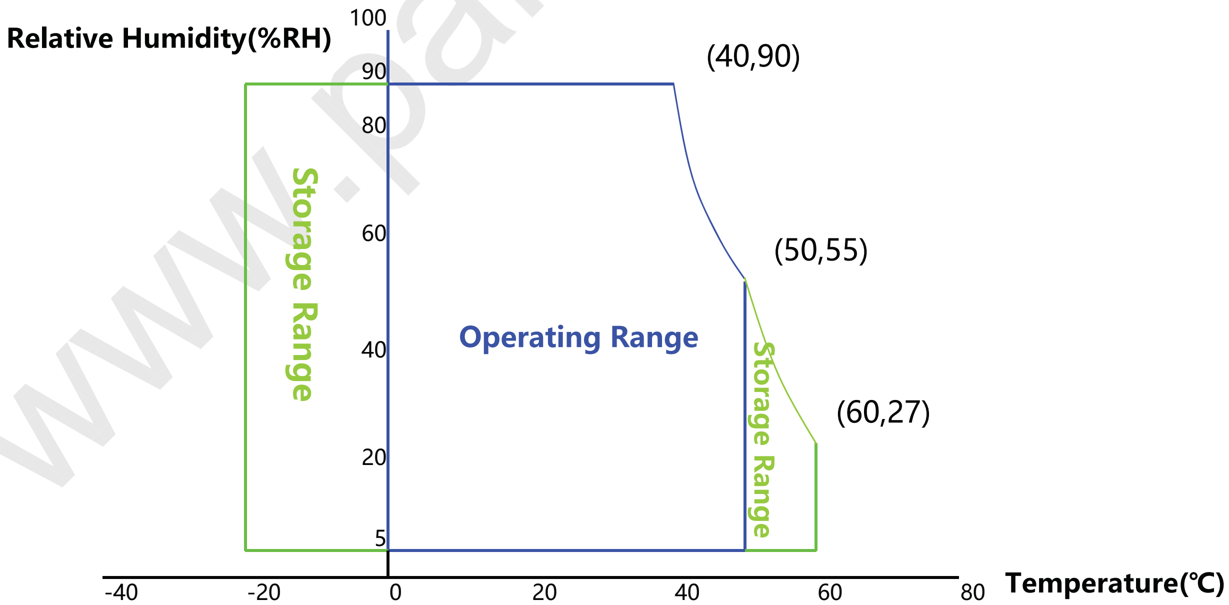


Figure 2. Temperature and Relative Humidity Range

3.0 ELECTRICAL SPECIFICATIONS

3.1 Electrical Specifications

< Table 3. Electrical Specifications > Ta=25+/-2°C

Parameter			Min.	Typ.	Max.	Unit	Remarks
Power Supply Voltage		V_{DD}	3.0	3.3	3.6	V	Note 1
Permissible Input Ripple Voltage		V_{RF}	-10% VDD	-	+10% VDD	V	@ V_{DD} = 3.3 V , note4
BIST Control Level		High Level	0.8 VDDIO	-	3.3	V	@VDDIO=1.8
		Low Level	0	-	0.15 VDDIO	V	
Power Supply Inrush Current		Inrush	-	-	2	A	Note3
Power Supply Current	Mosaic	I_{DD}	-	-	334	mA	Note 1
	RGB		-	-	364	mA	
Power Consumption	Mosaic	P_M	-	-	1.1	W	
	RGB	P_{RGB}	-	-	1.2	W	
	BLU	P_{BL}	-	-	4.82	W	Note 2
	Total	P_{Total}	-	-	5.92	W	@Mosaic

3.0 ELECTRICAL SPECIFICATIONS

3.1 Electrical Specifications

Notes :

1. The supply voltage is measured and specified at the interface connector of LCM.
   The current draw and power consumption specified is for 3.3V at 25 °C.
   a) Mosaic pattern 8*8
   b) R/G/B patterns
   c)Solid pattern(maximum logic power consumption) : sub v line

Figure 3. Power Measure Patterns

2. Calculated value for reference (VLED × ILED) , The power consumption with LED Driver are under the VLED = 12.0V , 25°C, PWM Duty 100% .

3. Measure condition (Figure 4)

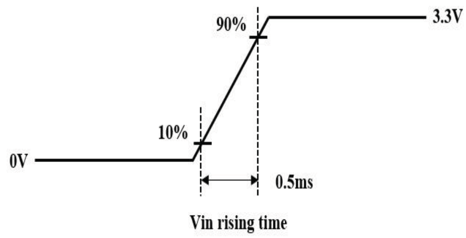


Figure 4. Inrush Measure Condition

4. Input voltage range:3.0~3.6V.Test condition: Oscilloscope bandwidth 20MHz, AC coupling

3.2 Backlight Unit

< Table 4. LED Driving Guideline Specifications > Ta=25+/-2°C

Parameter			Min.	Typ.	Max.	Unit	Remarks
LED Forward Voltage		V_F	-	-	2.9	V	
LED Forward Current		I_F	-	23	-	mA	
LED Power Input Voltage		VLED	5	12	21	V	
LED Power Input Current		I_{LED}	-	-	401.7	mA	Note 1
LED Power Consumption		P_{LED}	-	-	4.82	W	Note 1
Power Supply Voltage for LED Driver Inrush		Iled inrush	-	-	1.5	A	Note 3
LED Life-Time		N/A	15,000	-	-	Hour	IF = 23mA Note 2
EN Control Level	Backlight On	V_{BL_EN}	1.2	-	3.6	V	
	Backlight Off		0	-	0.6	V	
PWM Control Level	High Level	V_{BL_PWM}	1.2	-	3.6	V	
	Low Level		0	-	0.6	V	
PWM Control Frequency		F_{PWM}	200	-	10K	Hz	
Duty Ratio			5	-	100	%	

Notes :

1. Power supply voltage12V for LED driver.
   Calculator value for reference IF × VF ×60 /driver efficiency = PLED
2. The LED life-time define as the estimated time to 50% degradation of initial luminous.
3. Measure condition (Figure 5)

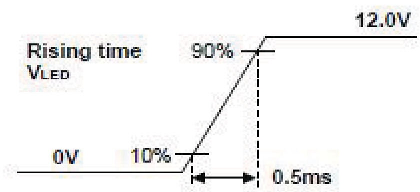


Figure 5. Inrush Measure Condition

3.3 LED Structure

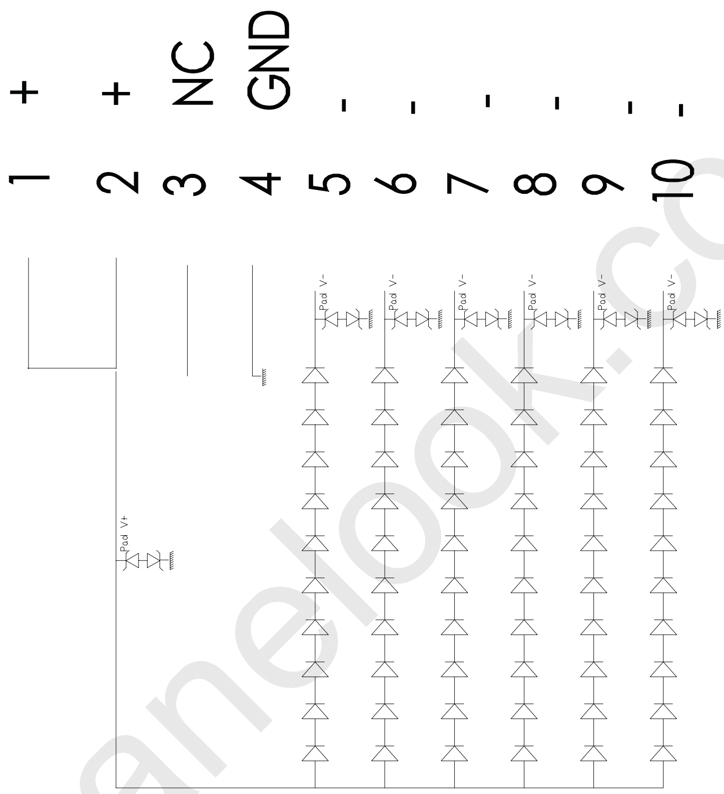


Figure 6. LED Structure

4.0 OPTICAL SPECIFICATION

4.1 Overview

The test of optical specifications shall be measured in a dark room (ambient luminance ≤ 1 lux and temperature = 25±2°C) with the equipment of luminance meter system (SR-3AR&GRT) and test unit shall be located at an approximate distance 50cm from the LCD surface at a viewing angle of θ and Φ equal to 0°. We refer to θØ=0 (=θ3) as the 3 o'clock direction (the "right"), θØ=90 (= θ12) as the 12 o'clock direction ("upward"), θØ=180 (= θ9) as the 9 o'clock direction ("left") and θØ=270(= θ6) as the 6 o'clock direction ("bottom"). While scanning θand/or Ø, the center of the measuring spot on the display surface shall stay fixed. The backlight should be operating for 30 minutes prior to measurement. VDD shall be 3.3+/- 0.3V at 25°C. Optimum viewing angle direction is 6 'clock.

4.2 Optical Specifications

<Table 5. Optical Specifications>

Parameter		Symbol	Condition	Min.	Typ.	Max.	Unit	Remark
Viewing Angle Range	Horizontal	Θ_3	CR > 10	80	89	-	Deg.	Note 1
		Θ_9		80	89	-	Deg.	
	Vertical	Θ_{12}		80	89	-	Deg.	
		Θ_6		80	89	-	Deg.	
Luminance Contrast Ratio		CR	$\Theta = 0°$	1200	1500	-		Note 2
Luminance of White	5 Points	Y_w	$\Theta = 0°$ I_{LED} =23mA	425	500	-	cd/m²	Note 3
White Luminance Uniformity	5 Points	ΔY5		80	-	-	%	Note 4
	13 Points	ΔY13		66.7	-	-	%	
White Chromaticity		W_x	$\Theta = 0°$	0.283	0.313	0.343		Note 5
		W_y		0.299	0.329	0.359		
Reproduction of Color	Red	R_x	$\Theta = 0°$	Typ.-0.03	0.680	Typ.+0.03		
		R_y			0.317			
	Green	G_x			0.259			
		G_y			0.695			
	Blue	B_x			0.144			
		B_y			0.060			
Color Gamut		DCI-P3		95	100	-	%	
Response Time (Rising + Falling)		T_{RT}	Ta= 25°C $\Theta = 0°$	-	25	30	ms	Note 6
Cross Talk		CT	$\Theta = 0°$	-	-	2.0	%	Note 7
Gamma		-	-	2.0	2.2	2.4	-	

Notes :

1. Viewing angle is the angle at which the contrast ratio is greater than 10. The viewing angles are determined for the horizontal or 3, 9 o'clock direction and the vertical or 6, 12 o'clock direction with respect to the optical axis which is normal to the LCD surface (see Figure 7).

2. Contrast measurements shall be made at viewing angle of Θ= 0 and at the center of the LCD surface. Luminance shall be measured with all pixels in the view field set first to white, then to the dark (black) state . (see Figure 7) Luminance Contrast Ratio (CR) is defined mathematically.

$$CR = \frac{\text{Luminance when displaying a white raster}}{\text{Luminance when displaying a black raster}}$$

3. Center Luminance of white is defined as luminance values of 5 point average across the LCD surface. Luminance shall be measured with all pixels in the view field set first to white. This measurement shall be taken at the locations shown in Figure 8 for a total of the measurements per display.

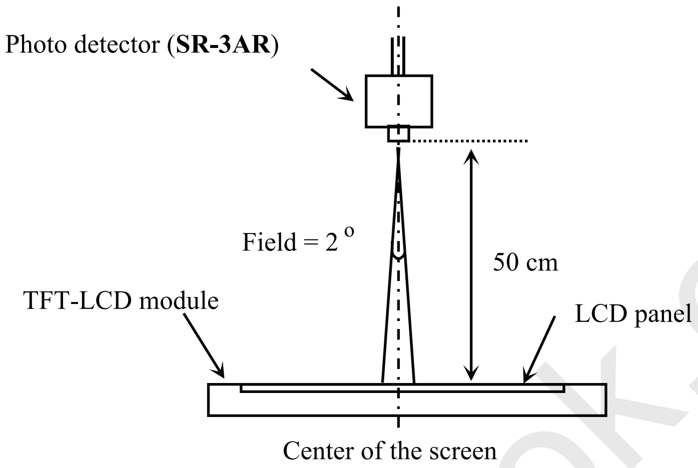
4. The White luminance uniformity on LCD surface is then expressed as : ΔY =Minimum Luminance of 5(or 13) points / Maximum Luminance of 5(or 13) points.(see Figure 8 and Figure 9).

5. The color chromaticity coordinates specified in Table 5 shall be calculated from the spectral data measured with all pixels first in red, green, blue and white. Measurements shall be made at the center of the panel.

6. The electro-optical response time measurements shall be made as Figure 10 by switching the "data" input signal ON and OFF. The times needed for the luminance to change from 10% to 90% is T_f, and 90% to 10% is T_r.

7. Cross-Talk of one area of the LCD surface by another shall be measured by comparing the luminance (YA) of a 25mm diameter area, with all display pixels set to a gray level, to the luminance (YB) of that same area when any adjacent area is driven dark. (See Figure 11).

4.3 Optical Measurements



Optical characteristics measurement setup

Figure 7. Measurement Set Up

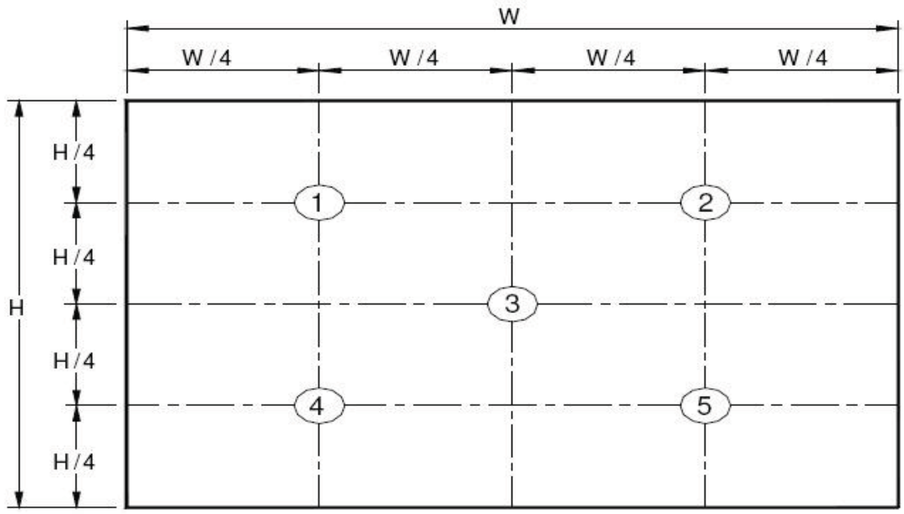


Figure 8. White Luminance and Uniformity Measurement Locations (5 points)

Center Luminance of white is defined as luminance values of center 5 points across the LCD surface. Luminance shall be measured with all pixels in the view field set first to white. This measurement shal l be taken at the locations shown in Figure 7 for a total of the measurements per display.

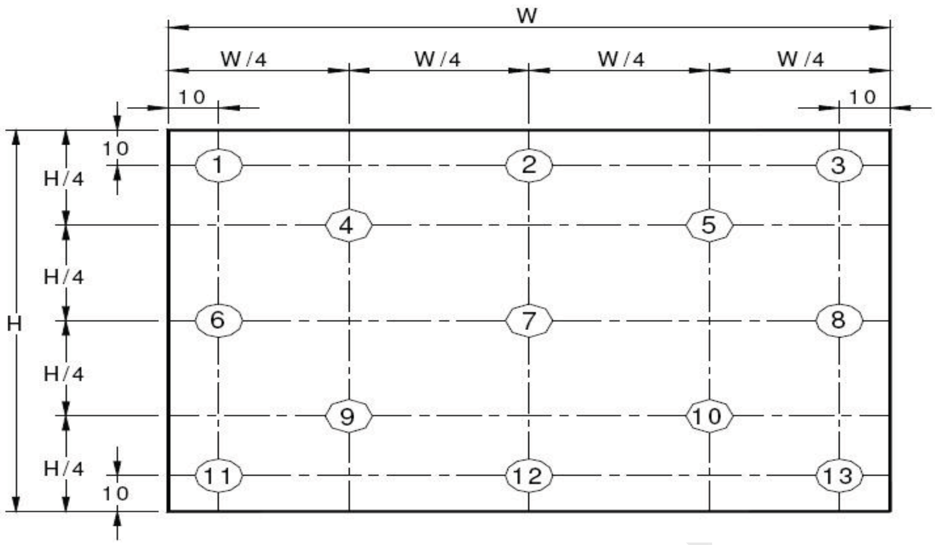


Figure 9. Uniformity Measurement Locations (13 points)

The White luminance uniformity on LCD surface is then expressed as : ΔY5 = Minimum Luminance of five points / Maximum Luminance of five points (see Figure 8) , ΔY13 = Minimum Luminance of 13 points /Maximum Luminance of 13 points (see Figure 9).

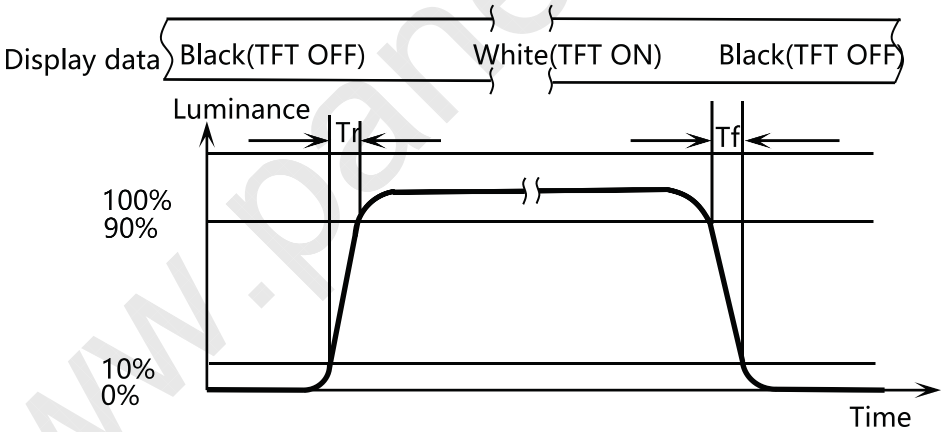
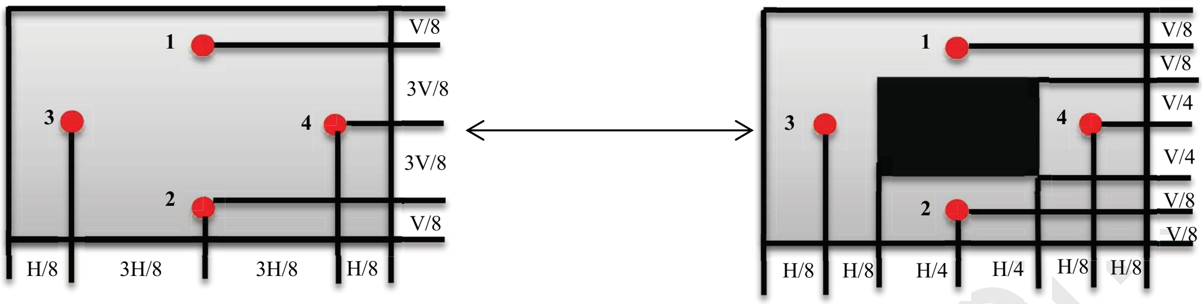


Figure 10. Response Time Testing

The electro-optical response time measurements shall be made as shown in Figure 10 by switching the “data” input signal ON and OFF. Tr: The luminance to change from 10% to 90% ,Tf: The luminance to change from 90% to 10% .

The test system : GRT



$$\text{Cross Talk (\%)} = \left| \frac{Y_B - Y_A}{Y_A} \right| \times 100$$

Figure 11. Cross Talk Modulation Test Description

Where:

Y_A = Initial luminance of measured area (cd/m²)

Y_B = Subsequent luminance of measured area (cd/m²)

The location 1/2/3/4 measured will be exactly the same in both patterns. The test background gray is from L64 to L192.Take the largest data as the result.

Cross Talk of one area of the LCD surface by another shall be measured by comparing the luminance (YA) of a 25mm diameter area, with all display pixels set to a gray level, to the luminance (YB) of tha t same area when any adjacent area is driven dark.(Refer to Figure 11)

The test system: SR-3AR

5.0 INTERFACE CONNECTION

5.1 Electrical Interface Connection

The electronics interface connector is I-PEX 20682-040E-2 or compitable.
The connector interface pin assignments are listed in Table 6.

<Table 6. Pin Assignments for the Interface Connector>

Terminal	Symbol	Functions
Pin No.	Symbol	Description
1	NC	No Connection
2	H_GND	Ground
3	LANE3_N	eDP RX Channel 3 Negative
4	LANE3_P	eDP RX Channel 3 Positive
5	H_GND	Ground
6	LANE2_N	eDP RX Channel 2 Negative
7	LANE2_P	eDP RX Channel 2 Positive
8	H_GND	Ground
9	LANE1_N	eDP RX Channel 1 Negative
10	LANE1_P	eDP RX Channel 1 Positive
11	H_GND	Ground
12	LANE0_N	eDP RX Channel 0 Negative
13	LANE0_P	eDP RX Channel 0 Positive
14	H_GND	Ground
15	AUX_CH_P	eDP AUX CH Positive
16	AUX_CH_N	eDP AUX CH Negative
17	H_GND	Ground
18	LCD_VCC	Power Supply, 3.3V (typ.)
19	LCD_VCC	Power Supply, 3.3V (typ.)
20	LCD_VCC	Power Supply, 3.3V (typ.)
21	LCD_VCC	Power Supply, 3.3V (typ.)
22	Bist	Bist
23	H_GND	Ground
24	H_GND	Ground
25	H_GND	Ground
26	H_GND	Ground
27	HPD	Hot Plug Detect Output
28	BL_GND	LED Ground
29	BL_GND	LED Ground
30	BL_GND	LED Ground

31	BL_GND	LED Ground
32	BL_ENABLE	LED Enable Pin(+3.3V Input)
33	BL_PWM	System PWM Signal Input
34	NC	No Connection
35	NC	No Connection
36	BL_POWER	LED Power Supply 5V-21V
37	BL_POWER	LED Power Supply 5V-21V
38	BL_POWER	LED Power Supply 5V-21V
39	BL_POWER	LED Power Supply 5V-21V
40	NC	No Connection

5.2 eDP Interface

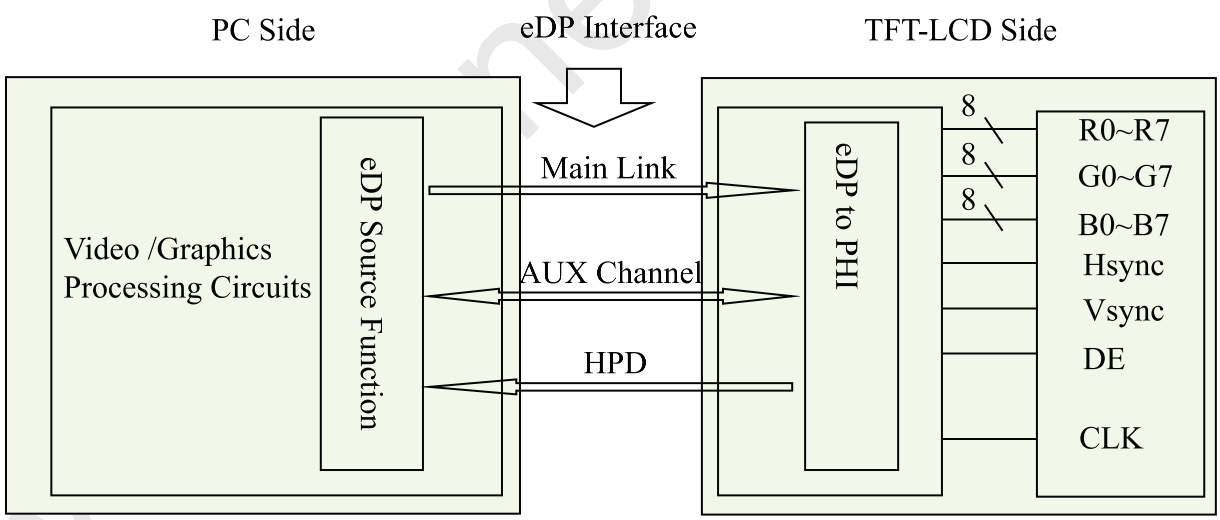


Figure 12. eDP Interface Architecture

Note:
Transmitter : Parade DP501 or equivalent.
Transmitter is not contained in module.

5.3 Data Input Format

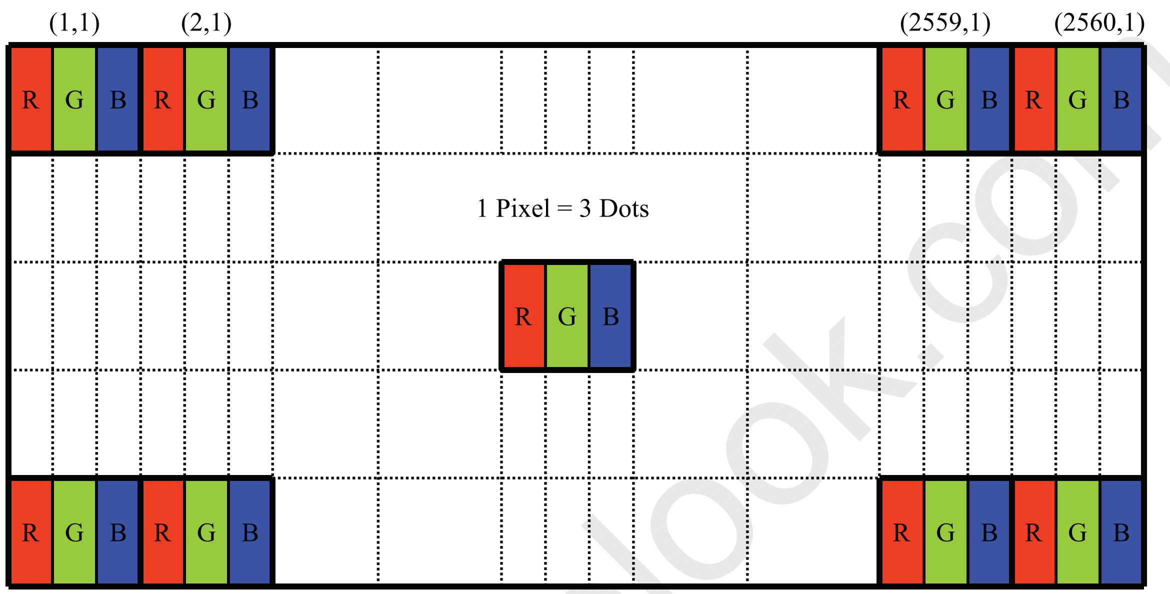


Figure 13. Display Position of Input Data (V-H)

5.4 Back-light & LCM Interface Connection

BLU Interface Connector: STM MSK24022P10 .

<Table 7. Pin Assignments for the BLU Connector>

Pin No.	Symbol	Description	Pin No.	Symbol	Description
1	Vout	LED anode connection	6	LED	LED cathode connection
2	Vout	LED anode connection	7	LED	LED cathode connection
3	NC	No Connection	8	LED	LED cathode connection
4	GND	Ground	9	LED	LED cathode connection
5	LED	LED cathode connection	10	LED	LED cathode connection

6.0 SIGNAL TIMING SPECIFICATION

6.1 The NE140QDM-NX5 Is Operated By The DE Only

< Table 8. Signal Timing Specification >

Item		Symbols	Min	Typ	Max	Unit
Clock	Frequency	1/Tc	-	554.88	-	MHz
Frame Period		Tv	-	1700	-	lines
			-	120	-	Hz
			-	8.33	-	ms
Vertical Display Period		Tvd	-	1600	-	lines
One line Scanning Period		Th	-	2720	-	clocks
Horizontal Display Period		Thd	-	2560	-	clocks

Note : The above is as optimized setting.

6.2 eDP Rx Interface Timing Parameter

The specification of the eDP Rx interface timing parameter is shown in Table 9.

<Table 9. eDP Main-Link RX TP4 Package Pin Parameters>

Item	Symbol	Min	Typ	Max	Unit	Remark
Spread spectrum clock (Link clock down-spreading)	ssc	-	-	0.5	%	
Differential peak-to-peak input voltage at package pins	VRX-DIFFp-p	120	-	1200	mV	
Rx input DC common mode voltage	VRX_DC_CM	0	-	2	V	
Differential termination resistance	RRX-DIFF	80	-	100	Ω	
Single-ended termination resistance	RRX-SE	40	-	60	Ω	
Rx short circuit current limit	IRX_SHORT	-	-	20	mA	
Intra-pair skew at Rx package pins (HBR) RX intra-pair skew tolerance at HBR	LRX_SKEW_INTRA_PAIR	-	-	60	ps	
AC Coupling Capacitor	CSOURCE_ML	75		200	nF	Source side

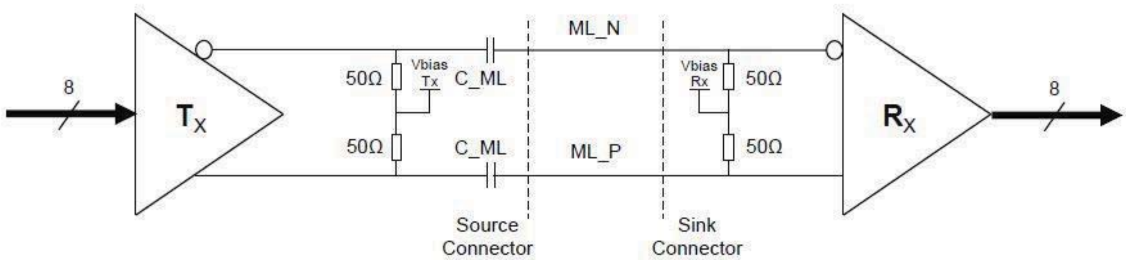


Figure 14. Main link differential pair

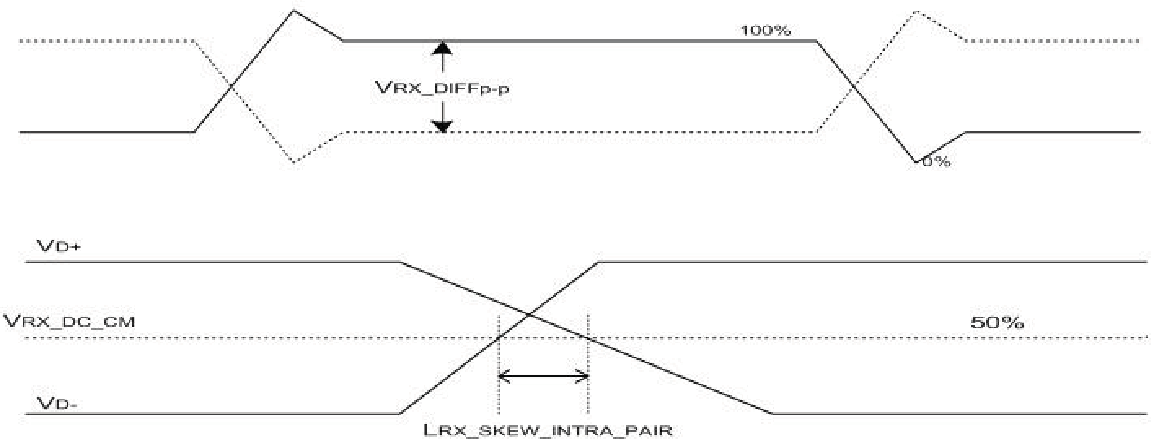


Figure 15. VRX-DIFFp-p & LRX_SKEW_INTRA_PAIR

<Table 10. HPD Characteristics>

Item	Symbol	Min	Typ	Max	Unit	Remark
HPD voltage	VHPD	2.25	-	3.6	V	
Hot Plug Detection Threshold	-	2.0	-	-	V	Source side Detecting
Hot Unplug Detection Threshold	-	-	-	0.8V	V	
HPD_IRQ Pulse Width	HPD_IRQ	0.5	-	1	ms	
HPD_TimeOut	-	2.0	-	-	ms	

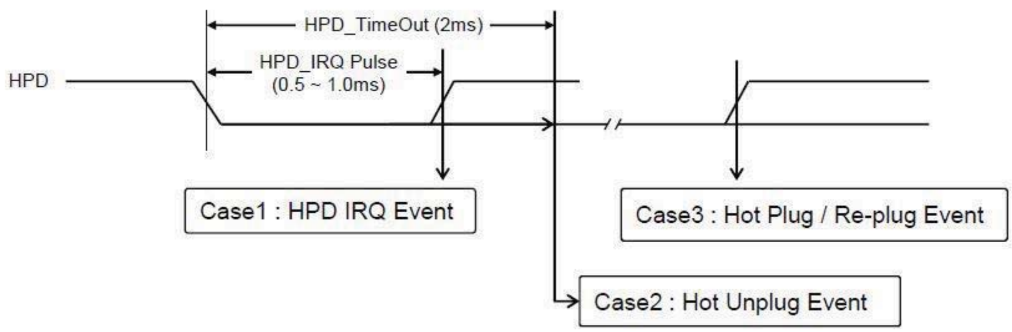


Figure 16. HPD Events

<Table 11. AUX Characteristics>

Item	Symbol	Min	Typ	Max	Unit	Remark
AUX unit interval	UIAUX	0.4	0.5	0.6	Us	
AUX peak-to-peak input differential voltage	VAUX-RX-DIFFp-p	0.29	-	1.38	V	
AUX CH termination DC resistance	RAUX-TERM	80	100	120	Ohm	
AUX DC common mode voltage	VAUX-DC-CM	0	-	2	V	
AUX turn around common mode voltage	VAUX-TURN-CM	-	-	0.3	V	
AUX short circuit current limit	IAUX-SHORT	-	-	90	mA	
AUX AC Coupling Capacitor	CSOURCE-AUX	75	-	200	nf	Source side

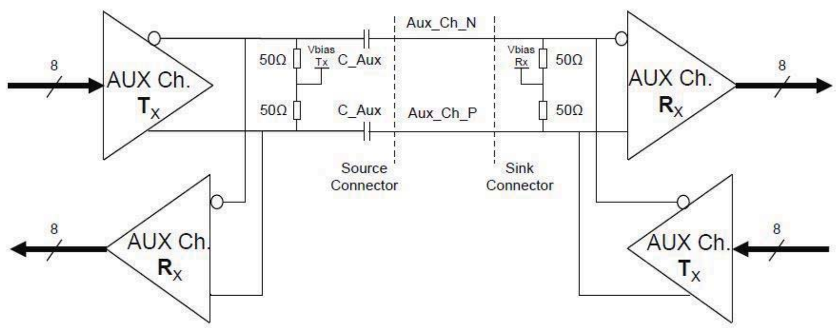


Figure 17. AUX differential pair

BOE	PRODUCT GROUP	REV	ISSUE DATE
	Customer Spec	Rev. P0	2024.02.19

7.0 INPUT SIGNALS, BASIC DISPLAY COLORS & GRAY SCALE OF COLORS

<Table 12. Input Signal & Basic Display Colors & Gray Scale of Colors >

	Colors & Gray scale	Data signal																							
		R0	R1	R2	R3	R4	R5	R6	R7	G0	G1	G2	G3	G4	G5	G6	G7	B0	B1	B2	B3	B4	B5	B6	B7
Basic colors	Black	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Blue	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1
	Green	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0
	Light Blue	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	Red	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Purple	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1
	Yellow	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0
	White	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Gray scale of Red	Black	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	▵	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Darker	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	▵				↑								↑								↑				
	▿				↓								↓								↓				
	Brighter	1	0	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	▿	0	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Red	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Gray scale of Green	Black	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	▵	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Darker	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	▵				↑								↑								↑				
	▿				↓								↓								↓				
	Brighter	0	0	0	0	0	0	0	0	1	0	1	1	1	1	1	1	0	0	0	0	0	0	0	0
	▿	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0
	Green	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0
Gray scale of Blue	Black	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	▵	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
	Darker	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0
	▵				↑								↑								↑				
	▿				↓								↓								↓				
	Brighter	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	1	1	1	1
	▿	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1
	Blue	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1
Gray scale of White& Black	Black	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	▵	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
	Darker	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0
	▵				↑								↑								↑				
	▿				↓								↓								↓				
	Brighter	1	0	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	0	1	1	1	1	1	1
	▿	0	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1
	White	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

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8.0 POWER SEQUENCE

To prevent a latch-up or DC operation of the LCD module, the power on/off sequence shall be as shown in below.

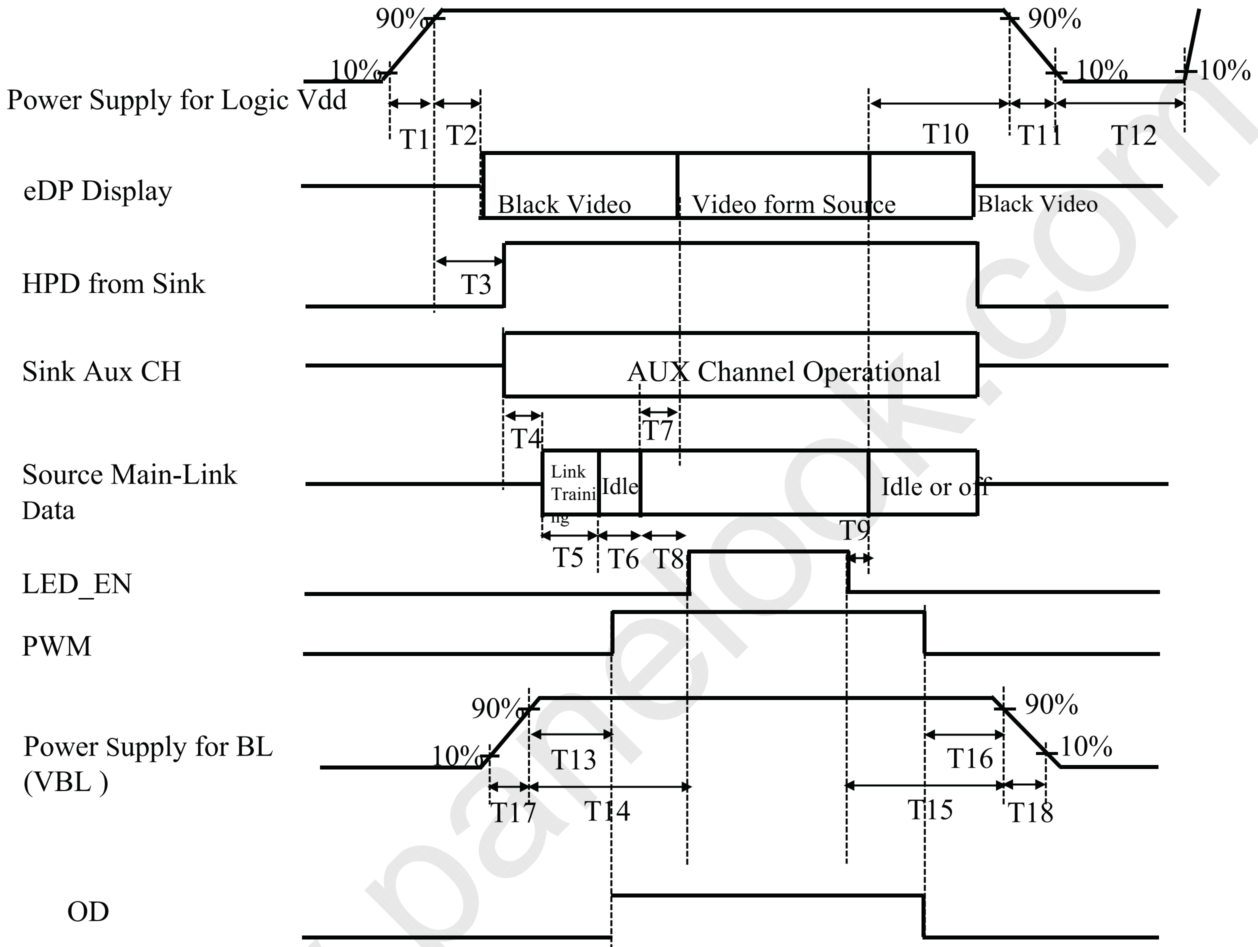
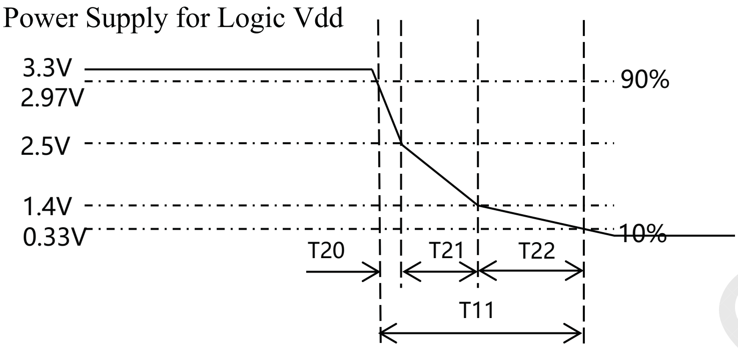


Figure 18. Power Sequence

- 0.5ms ≤ T1 ≤ 10 ms
- 0ms < T2 ≤ 200 ms
- 0ms < T3 ≤ 200 ms
- T4+T5+T6+T8>80ms
- 0ms < T7 ≤ 50ms
- 50ms < T8
- -100ms < T9
- 100ms < T10 < 500 ms
- 0.5ms ≤ T11 ≤ 10 ms
- 500ms ≤ T12
- 0ms < T13
- 0ms < T14
- 0ms < T15
- 0ms < T16
- 0.5ms ≤ T17
- 0.5ms ≤ T18

Notes:

1. When the power supply VDD is 0V, keep the level of input signals on the low or keep high impedance.
2. Do not keep the interface signal high impedance when power is on. Back Light must be turn on after power for logic and interface signal are valid.



● 0.5ms ≤ T11 ≤ 10 ms ● 0.225ms ≤ T21 ● T11=T20+T21+T22

Figure 19. T11 timing requirements

9.0 Connector Description

Physical interface is described as for the connector on LCM.
These connectors are capable of accommodating the following signals and will be following components.

9.1 TFT LCD Module

< Table 13. Signal Connector >

Connector Name /Description	For Signal Connector
Manufacturer	I-PEX
Type/ Part Number	20682-040E-02
Mating Housing/ Part Number	20856-040T-01

10.0 MECHANICAL CHARACTERISTICS

10.1 Dimensional Requirements

Figure 23 shows mechanical outlines for the model NE140QDM-NX5.
Other parameters are shown in Table 14.

<Table 14. Dimensional Parameters>

Parameter	Specification	Unit
Active Area	301.59(H) ×188.50(V)	mm
Number of pixels	2560 (H) X 1600 (V) (1 pixel = R + G + B dots)	pixels
Pixel pitch	117.81(H) ×117.81(V)	um
Pixel arrangement	RGB Vertical stripe	
Display colors	1.07B(8bit+FRC)	
Display mode	Normally black	
Dimensional outline	306.59±0.3 (H)*198.4±0.5(V)*1.95(Max)(W/O PCB) 306.59±0.3(H)*198.4±0.5(V) *3.95(Max)(W/PCB)	mm
Weight	180 (max)	g

10.2 Mounting

See Figure 24.

10.3 Anti-Glare and Polarizer Hardness.

The surface of the LCD has an Anti-Glare coating to minimize reflection and a coating to reduce scratching.

10.4 Light Leakage

There shall not be visible light from the back-lighting system around the edges of the screen as seen fro m a distance 50cm from the screen with an overhead light level of 350lux.

11.0 RELIABILITY TEST

The reliability test items and its conditions are shown in below.

<Table 15. Reliability Test>

No	Test Items	Conditions	Remark
1	High temperature storage test	Ta = 60°C , 240 hrs	
2	Low temperature storage test	Ta = -20°C , 240 hrs	
3	High temperature & high hu midity operation test	Ta = 50°C , 80%RH, 240 hrs	
4	High temperature operation te st	Ta = 50°C , 240 hrs	
5	Low temperature operation te st	Ta = 0°C , 240 hrs	
6	Thermal shock	Ta = -20 °C(0.5 hr), ↔ 60 °C (0.5 hr), Switc h time 5min , 100 cycle	
7	Vibration test (non-operating)	Ta = 25°C , 60%RH, 1.5G, 10~500Hz, Sine wave, 60min/cycle, 1cycle for each X,Y, Z	Note 1
8	Shock test (non-operating)	Ta = 25°C , 60%RH, 220G, Half Sine Wave 2 msec±X,±Y,±Z Once for each direction	Note 1
9	Electro-static discharge test (operating)	Air : 150 pF, 330Ω, ±15 KV Contact : 150 pF, 330Ω, ±8 KV Ta = 25°C , 60%RH,	Note 2

Notes :

1. The fixture must be hard enough , so that the module would not be twisted or bent.
2. Self- recovery and restart recovery is allowed. No hardware failures.

12.0 HANDLING & CAUTIONS

(1) Cautions when taking out the module
- Pick the pouch only, when taking out module from a shipping package.

(2) Cautions for handling the module
- As the electrostatic discharges may break the LCD module, handle the LCD module with care. Peel a protection sheet off from the LCD panel surface as slowly as possible.
- As the LCD panel and back - light element are made from fragile glass material, impulse and pressure to the LCD module should be avoided.
- As the surface of the polarizer is very soft and easily scratched, use a soft dry cloth without chemicals for cleaning.
- Do not pull the interface connector in or out while the LCD module is operating.
- Put the module display side down on a flat horizontal plane.
- Handle connectors and cables with care.

(3) Cautions for the operation
- When the module is operating, do not lose CLK, ENAB signals. If any one of these signals is lost, the LCD panel would be damaged.
- Obey the supply voltage sequence. If wrong sequence is applied, the module would be damaged.

(4) Cautions for the atmosphere
- Dew drop atmosphere should be avoided.
- Do not store and/or operate the LCD module in a high temperature and/or humidity atmosphere. Storage in an electro-conductive polymer packing pouch and under relatively low temperature atmosphere is recommended.

(5) Cautions for the module characteristics
- Do not apply fixed pattern data signal to the LCD module at product aging.
- Applying fixed pattern for a long time may cause image sticking.

(6) Other cautions
- Do not disassemble and/or re-assemble LCD module.
- Do not re-adjust variable resistor or switch etc.
- When returning the module for repair or etc. Please pack the module not to be broken. We recommend to use the original shipping packages.

13.0 LABEL

(1) Product Label

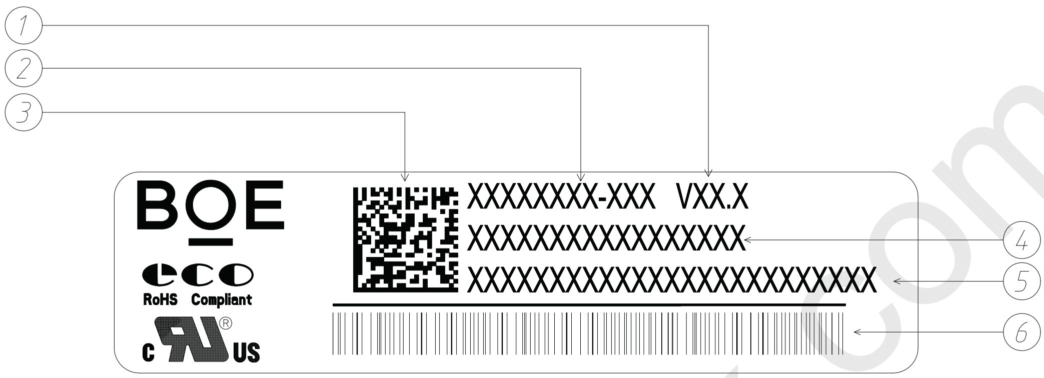


Figure 22. Product Label

Serial number marked part needs to print, show as follows:

1. Version
2. FG-CODE (Before 12 bit)
3. ASUS CC ID (2D barcode)
4. Module ID
5. ASUS CC ID
6. Module ID barcode

Module ID Naming Rule:

<Table 16. Module ID Naming Rule>

Digit / Code	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Code	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Description	Product Name		Product Grade	Facility Code	Year		Month	Model Extension Code (Last 4 Digits FG Code)				Serial NO.					

(2) Box label

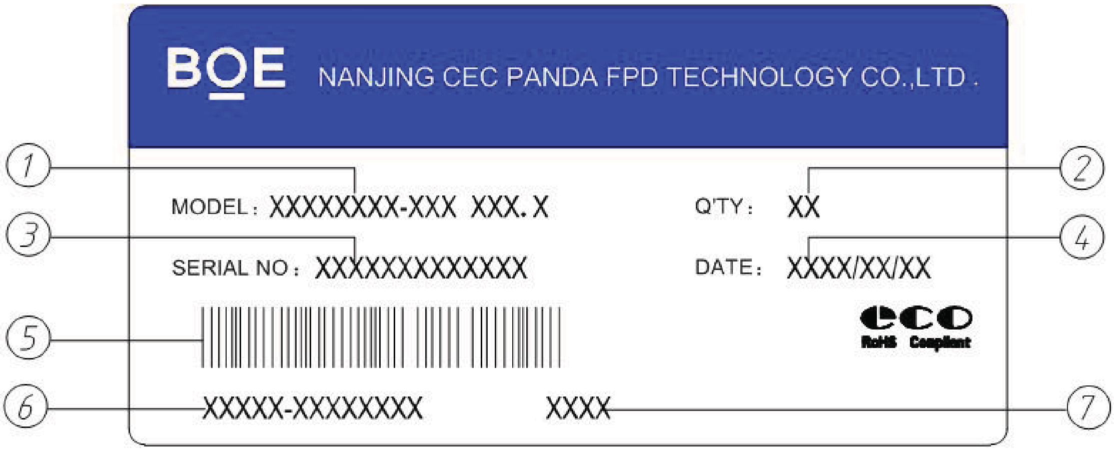


Figure 22. Box Label

Packing bar code label:

1.FG-Code (Before 12 bit)+V18.3

2.Product Quantity(XX pcs/Carton)

3.BOX ID

4.Date of Packing

5.BOX ID Serial number Barcode

6. Customer NO

7.FG-Code After four

Label size：100*50mm

Box ID Naming Rule:

<Table 17. Box Label Naming Rule >

Digit / Code	1	2	3	4	5	6	7	8	9	10	11	12	13
Code	X	X	X	X	X	X	X	X	X	X	X	X	X
Description	Product Name		Product Grade	Facility Code	Year		Month	Revision	Box Serial NO.				

14.0 PACKING INFORMATION

14.1 Packing Order

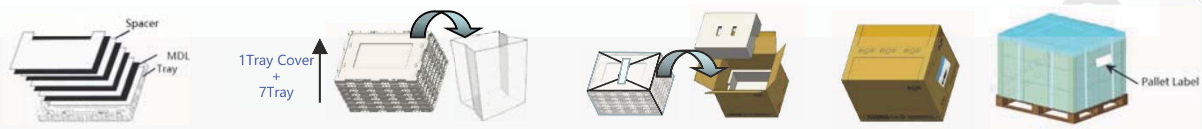


Figure 21. Packing Order

- Put 1pcs EPE Spacer in Tray and 1pcs MDL on Spacer.

  Capacity:4pcs MDL/Tray , 5pcs Spacer/Tray
- Put 7 pcs tray and 1 pcs tray cover in PE bag.
- Put PE bag with 2pcs EPE cover in the inner box.

  Capacity:28pcs MDL/Box,
- Use Packing Tape to Seal inner box,Stick the box label onto inner box and align it to the label mark.
- Put 18EA Box on the Pallet , Secure with strapping tape, wrap around film, paper protection Angle.

  Capacity:6EA Box/Layer, 3Layer, 504pcs MDL/Pallet

14.2 Note

- Box dimension: 472mm*352mm*294mm
- Package quantity in one box: 28pcs
- Total weight: （9.06±10%） kg/Box

15.0 MECHANICAL OUTLINE DIMENSION

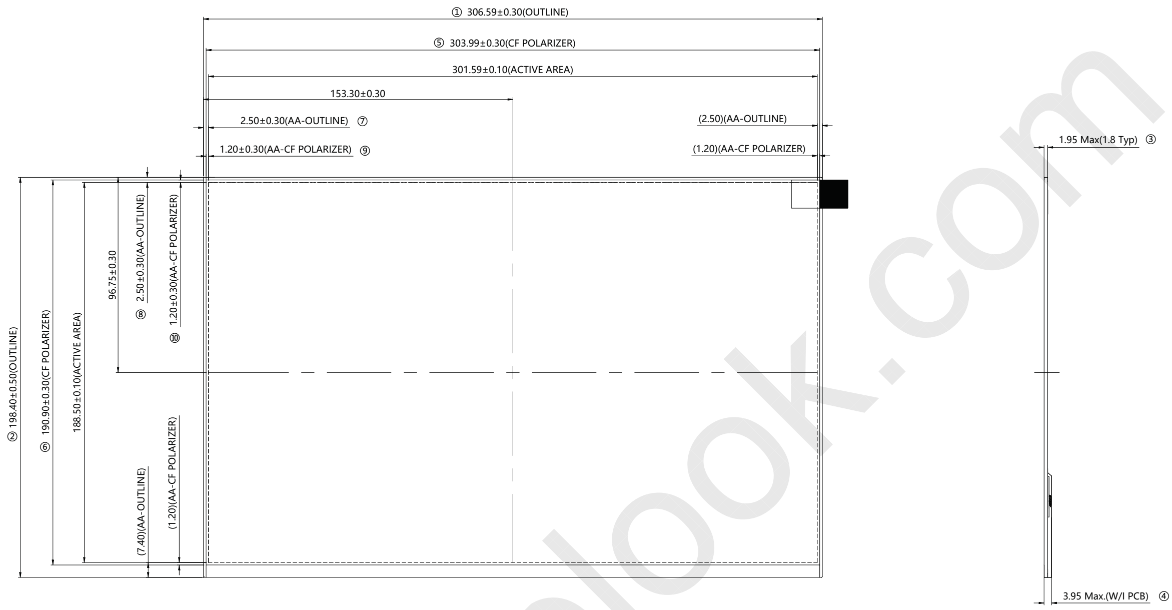


Figure 24. TFT-LCD Module Outline Dimension (Front View)

NOTES:

1.THE eDP CONNECTOR IS MEASURED AT PIN 1 AND MATING LINE

2.UNSPECIFIED TOLERANCE REFER TO ±0.3 mm

3.THE MEASUREMENT METHOD FOR THE DIMENSION OF MODULE, PLEASE REFR E TO PRODUCT SPEC..

4.TOP POL IS THE HIGHEST POSITION

5."()" MEANS REFERANCE DIMENSIONS.

6.CRITICAL DIMENSION: ①~⑭

CPK ：①②

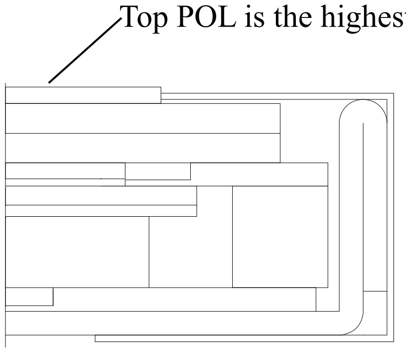


Figure 25. Highest Point Position

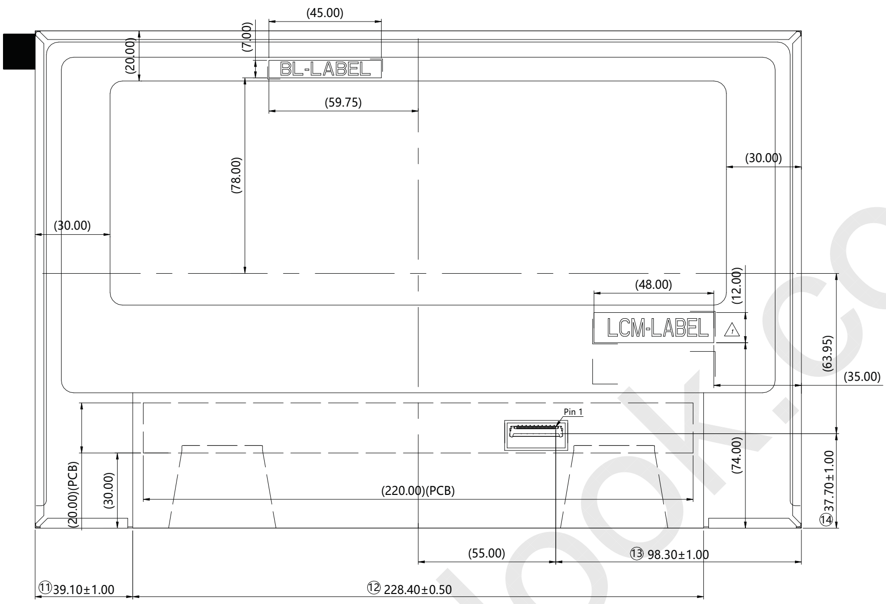


Figure 26. TFT-LCD Module Outline Dimensions (Rear view)

NOTES:
1.THE eDP CONNECTOR IS MEASURED AT PIN 1 AND MATING LINE
2.UNSPECIFIED TOLERANCE REFER TO ±0.3 mm
3.THE MEASUREMENT METHOD FOR THE DIMENSION OF MODULE, PLEASE REFR E TO PRODUCT SPEC..
4.TOP POL IS THE HIGHEST POSITION
5."()" MEANS REFERANCE DIMENSIONS.
6.CRITICAL DIMENSION: ①~⑭
CPK：①②

16.0 EDID Table

Check FAE	Check QE	Address (HEX)	Function	Hex	Dec	crc	Input values.	Notes
-	-	00	Header	00	0		0	EDID Header
-	-	01		FF	255		255	
-	-	02		FF	255		255	
-	-	03		FF	255		255	
-	-	04		FF	255		255	
-	-	05		FF	255		255	
-	-	06		FF	255		255	
-	-	07		00	0		0	
V		08	ID Manufacturer Name	09	9		BOE	ID = BOE
V		09		E5	229			
	V	0A	ID Product Code	1B	27		3099	ID = 3099
	V	0B		0C	12			
V		0C	32-bit serial No.	00	0		0	
V		0D		00	0		0	
V		0E		00	0		0	
V		0F		00	0		0	
V		10	Week of manufacture	12	18		18	
V		11	Year of Manufacture	21	33		2023	Manufactured in 2023
V		12	EDID Structure Ver.	01	1		1	EDID Ver 1.0
V		13	EDID revision #	04	4		4	EDID Rev. 0.4
V	V	14	Video input definition	B5	181		-	Refer to right table
	V	15	Max H image size	1E	30		30	30.1594 cm (Approx)
	V	16	Max V image size	13	19		19	18.8496 cm (Approx)
	V	17	Display Gamma	78	120		2.2	Gamma curve = 2.2
V		18	Feature support	03	3		-	Refer to right table
	V	19	Red/Green low bits	14	20		-	Red / Green Low Bits
	V	1A	Blue/White low bits	D5	213		-	Blue / White Low Bits
	V	1B	Red x high bits	AE	174	696	0.680	Red (x) = 10101110 (0.68)
	V	1C	Red y high bits	51	81	325	0.317	Red (y) = 01010001 (0.317)
	V	1D	Green x high bits	42	66	265	0.259	Green (x) = 01000010 (0.259)
	V	1E	Green y high bits	B2	178	712	0.695	Green (y) = 10110010 (0.695)
	V	1F	Blue x high bits	24	36	147	0.144	Blue (x) = 00100100 (0.144)
	V	20	BLue y high bits	0F	15	61	0.060	Blue (y) = 00001111 (0.06)
	V	21	White x high bits	50	80	321	0.313	White (x) = 01010000 (0.313)
	V	22	White y high bits	54	84	337	0.329	White (y) = 01010100 (0.329)
V		23	Established timing 1	00	0		-	Refer to right table
V		24	Established timing 2	00	0		-	
V		25	Established timing 3	00	0		-	

V		26	Standard timing #1	01	1			Not Used
V		27		01	1			
V		28	Standard timing #2	01	1			Not Used
V		29		01	1			
V		2A	Standard timing #3	01	1			Not Used
V		2B		01	1			
V		2C	Standard timing #4	01	1			Not Used
V		2D		01	1			
V		2E	Standard timing #5	01	1			Not Used
V		2F		01	1			
V		30	Standard timing #6	01	1			Not Used
V		31		01	1			
V		32	Standard timing #7	01	1			Not Used
V		33		01	1			
V		34	Standard timing #8	01	1			Not Used
V		35		01	1			
	V	36	Detailed timing/monitor descriptor #1	C0	192		554.880	554.88MHz Main clock
	V	37		D8	216			
	V	38		00	0		2560	Hor Active = 2560
	V	39		A0	160		160	Hor Blanking = 160
	V	3A		A0	160		-	4 bits of Hor. Active + 4 bits of Hor. Blanking
	V	3B		40	64		1600	Ver Active = 1600
	V	3C		64	100		100	Ver Blanking = 100
	V	3D		60	96		-	4 bits of Ver. Active + 4 bits of Ver. Blanking
	V	3E		30	48		48	Hor Sync Offset = 48
	V	3F		20	32		32	H Sync Pulse Width = 32
	V	40		36	54		3	V sync Offset = 3 line
	V	41		00	0		6	V Sync Pulse width : 6 line
	V	42		2E	46		302	Horizontal Image Size = 301.594 mm (Low 8 bits)
	V	43		BC	188		188	Vertical Image Size = 188.496 mm (Low 8 bits)
	V	44		10	16		-	4 bits of Hor Image Size + 4 bits of Ver Image Size
	V	45		00	0		0	Hor Border (pixels)
	V	46		00	0		0	Vertical Border (Lines)
	V	47		1A	26		-	Refer to right table

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V		48	Detailed timing/monitor descriptor #2	60	96		277.440	277.44MHz Main clock
V		49		6C	108			
V		4A		00	0		2560	Hor Active = 2560
V		4B		A0	160		160	Hor Blanking = 160
V		4C		A0	160		-	4 bits of Hor. Active + 4 bits of Hor. Blanking
V		4D		40	64		1600	Ver Active = 1600
V		4E		64	100		100	Ver Blanking = 100
V		4F		60	96		-	4 bits of Ver. Active + 4 bits of Ver. Blanking
V		50		30	48		48	Hor Sync Offset = 48
V		51		20	32		32	H Sync Pulse Width = 32
V		52		36	54		3	V sync Offset = 3 line
V		53		00	0		6	V Sync Pulse width : 6 line
V		54		2E	46		302	Horizontal Image Size = 301.594 mm (Low 8 bits)
V		55		BC	188		188	Vertical Image Size = 188.496 mm (Low 8 bits)
V		56		10	16		-	4 bits of Hor Image Size + 4 bits of Ver Image Size
V		57		00	0		0	Hor Border (pixels)
V		58		00	0		0	Vertical Border (Lines)
V		59		1A	26		-	Refer to right table
V		5A	Detailed timing/monitor descriptor #3	00	0		-	Indicates descriptor is a display Descriptor
V		5B		00	0		-	
V		5C		00	0		-	Reserved
V		5D		FD	253		-	Tag Number for Display Range Limits Descriptor
V		5E		00	0		0	Vertical/Horizontal Rate Offset
V		5F		30	48		48	Minimum Vertical Rate:48 Hz
V		60		78	120		120	Maximum Vertical Rate:120 Hz
V		61		CC	204		204.0	Minimum Horizontal Rate:204 kHz
V		62		CC	204		204.0	Maximum Horizontal Rate:204 kHz
V		63		37	55		554.880	Maximum Pixel Clock:554.88 MHz
V		64		01	1		-	Range Limits Only
V		65		0A	10		-	Display Range Limits & CVT Support Definition Format terminate with ASCII code 0Ah and pad field with ASCII code 20h
V		66		20	32		-	
V		67		20	32		-	
V		68		20	32		-	
V		69		20	32		-	
V		6A		20	32		-	
V		6B		20	32		-	

V		6C	Detailed timing/monitor descriptor #4	00	0			Indicates descriptor #4 is a display Descriptor
V		6D		00	0			
V		6E		00	0			Reserved
V		6F		FE	254			Tag : ASCII String
V		70		00	0			Reserved
V		71		4E	78		N	Model name : NE140QDM-NX5
V		72		45	69		E	
V		73		31	49		1	
V		74		34	52		4	
V		75		30	48		0	
V		76		51	81		Q	
V		77		44	68		D	
V		78		4D	77		M	
V		79		2D	45		-	
V		7A		4E	78		N	
V		7B		58	88		X	
V		7C		35	53		5	
V		7D		0A	10			
V		7E	Extension flag	01	1		1	0 ：1個EDID；N-1：N个EDID
V		7F	Checksum	1B	27	27	-	
V		80	Tag	02	2		2	Tag 0x02 assigned by VESA to CTA for this extension
V		81	Revision Number	03	3		3	indicates revision of CTA-861 used by this device (Ver3)
V		82	Length of Info Frame	22	34		34	Byte number offset where 18-byte descriptors begin (typically Detailed Timing Descriptors)
V		83	Global Declarations	00	0		0	these bit are utilized as flags in later versions of CTA-861
V		84		E3	227		227	Colorimetry Data Block , 3 Byte
V		85		05	5		5	Colorimetry Data Block
V		86		80	128		128	support BT2020RGB
V		87		00	00		00	128：DCI-P3 panel
V		88		E6	230		230	Colorimetry Data Block , 6 Byte
V		89		06	6		6	HDR Static Metadata Data Block
V		8A		05	5		5	Traditional gamma SDR, SMPTE ST 2084 [2]
V		8B		01	1		1	Static Metadata Type 1
V		8C		6A	106		106	panel real typ luminance 500
V		8D		6A	106		106	panel real typ luminance 500
V		8E		43	67		67	panel real min luminance 0.35
V		8F		72	114		114	CEA Data Block for AMD VSDB3=114, VSDB2=109

V		90	AMD IEEE OUI Value	1A	26		26	24bit AMD IEEE OUI Value (0x00001A) (least significant byte first) PB1 = 0x1A, PB2 = 0x00, PB3 = 0x00
V		91		00	0			
V		92		00	0			
V		93	AMD VSDB Version	03	3		3	AMD VSDB Version 3
V		94	FreeSync Capability	01	1		1	
V		95	Min Refresh Rate	30	48		48	Min Refresh Rate in 48 Hz
V		96	Max Refresh Rate	78	120		120	Max Refresh Rate in 120 Hz
V		97	VCP code	00	0		0	0x00 or MCCS VCP code to query current state of FreeSync
V		98		00	00		0	Not supports seamless Gamma 2.2 transfer curve switching
V		99		6A	106		106	Max Luminance 1 (for HDR) = 500 Cd/m2
V		9A		43	67		67	Max Luminance 1 (for HDR) = 0.35 Cd/m2
V		9B		6A	106		106	Max Luminance 1 (for HDR) = 500 Cd/m2
V		9C		43	67		67	Max Luminance 1 (for HDR) = 0.35 Cd/m2
V		9D		78	120		120	Bits 7:0-LSB Freesync Maximun refresh rate[Hz]
V		9E		00	0		0	Max Refresh Rate>255Hz Reserved Bits 9:8-MSB Freesync Maximun refresh rate[Hz]
V		9F		00	0		0.00	Pixel Clock (Low bit, Range = 0.001Mhz (000000h) ~ 16,777.216Mhz (FFFFFFh))
V		A0		00	0			Maximum Pixel Clock (Middle bit)
V		A1		00	0			Maximum Pixel Clock (High bit)
V		A2		00	0		-	Unused
V		A3		00	0		-	Unused
V		A4		00	0		-	Unused
V		A5		00	0		-	Unused
V		A6		00	0		-	Unused
V		A7		00	0		-	Unused
V		A8		00	0		-	Unused
V		A9		00	0		-	Unused
V		AA		00	0		-	Unused
V		AB		00	0		-	Unused
V		AC		00	0		-	Unused
V		AD		00	0		-	Unused
V		AE		00	0		-	Unused
V		AF		00	0		-	Unused
V		B0		00	0		-	Unused
V		B1		00	0		-	Unused
V		B2		00	0		-	Unused
V		B3		00	0		-	Unused

V		B4		00	0		-	Unused
V		B5		00	0		-	Unused
V		B6		00	0		-	Unused
V		B7		00	0		-	Unused
V		B8		00	0		-	Unused
V		B9		00	0		-	Unused
V		BA		00	0		-	Unused
V		BB		00	0		-	Unused
V		BC		00	0		-	Unused
V		BD		00	0		-	Unused
V		BE		00	0		-	Unused
V		BF		00	0		-	Unused
V		C0		00	0		-	Unused
V		C1		00	0		-	Unused
V		C2		00	0		-	Unused
V		C3		00	0		-	Unused
V		C4		00	0		-	Unused
V		C5		00	0		-	Unused
V		C6		00	0		-	Unused
V		C7		00	0		-	Unused
V		C8		00	0		-	Unused
V		C9		00	0		-	Unused
V		CA		00	0		-	Unused
V		CB		00	0		-	Unused
V		CC		00	0		-	Unused
V		CD		00	0		-	Unused
V		CE		00	0		-	Unused
V		CF		00	0		-	Unused
V		D0		00	0		-	Unused
V		D1		00	0		-	Unused
V		D2		00	0		-	Unused
V		D3		00	0		-	Unused
V		D4		00	0		-	Unused
V		D5		00	0		-	Unused
V		D6		00	0		-	Unused
V		D7		00	0		-	Unused

V		D8		00	0		-	Unused
V		D9		00	0		-	Unused
V		DA		00	0		-	Unused
V		DB		00	0		-	Unused
V		DC		00	0		-	Unused
V		DD		00	0		-	Unused
V		DE		00	0		-	Unused
V		DF		00	0		-	Unused
V		E0		00	0		-	Unused
V		E1		00	0		-	Unused
V		E2		00	0		-	Unused
V		E3		00	0		-	Unused
V		E4		00	0		-	Unused
V		E5		00	0		-	Unused
V		E6		00	0		-	Unused
V		E7		00	0		-	Unused
V		E8		00	0		-	Unused
V		E9		00	0		-	Unused
V		EA		00	0		-	Unused
V		EB		00	0		-	Unused
V		EC		00	0		-	Unused
V		ED		00	0		-	Unused
V		EE		00	0		-	Unused
V		EF		00	0		-	Unused
V		F0		00	0		-	Unused
V		F1		00	0		-	Unused
V		F2		00	0		-	Unused
V		F3		00	0		-	Unused
V		F4		00	0		-	Unused
V		F5		00	0		-	Unused
V		F6		00	0		-	Unused
V		F7		00	0		-	Unused
V		F8		00	0		-	Unused
V		F9		00	0		-	Unused
V		FA		00	0		-	Unused
V		FB		00	0		-	Unused

V		F9		00	0		-	Unused
V		FA		00	0		-	Unused
V		FB		00	0		-	Unused
V		FC		00	0		-	Unused
V		FD		00	0		-	Unused
V		FE	Checksum(81~FD)	00	0		-	Unused
V		FF	Checksum(80~FE)	5E	94			

17.0 GENERAL PRECAUTIONS

17.1 HANDLING

(1) When the module is assembled, It should be attached to the system firmly using every mounting holes. Be careful not to twist or bend the modules.

(2) Refrain from strong mechanical shock or any force to the module. Otherwise, it may cause improper operation or damage to the module.

(3) Note that polarizers are very fragile and could be easily damaged. Do not press or scratch the surface harder than 1 HB pencil lead.

(4) Wipe off water droplets or oil immediately. If you leave the droplets for a long time, Staining and discoloration may occur.

(5) If the surface of the polarizer is dirty, clean it using some absorbent cotton or soft cloth.

(6) The desirable cleaners are water, IPA (Isopropyl Alcohol) or Hexane. Do not use Ketone type materials(ex. Acetone), Ethyl alcohol, Toluene, Ethyl acid or Methyl chloride. It might permanently damage to the polarizer due to chemical reaction.

(7) If the liquid crystal material leaks from the panel, it should be kept away from the eyes or mouth .In case of contact with hands, legs or clothes, it must be washed away thoroughly with soap.

(8) Protect the module from static , it may cause damage to the module.

(9) Use fingerstalls with soft gloves to keep display clean during the incoming inspection and assembly process.

(10) Do not disassemble the module.

(11) Do not pull or fold the LED FPC.

(l2) Do not touch any component which is located on the back side.

(13) Protection film for polarizer on the module shall be slowly peeled off just before use so that the electrostatic charge can be minimized.

(14) Pins of connector shall not be touched directly with bare hands.

17.2 STORAGE

(1) Do not leave the module in high temperature, and high humidity for a long time. It is highly recommended to store the module with temperature from 0 to 35°C and relative humidity of less than 70%.

(2) Do not store the TFT-LCD module in direct sunlight.

(3) The module shall be stored in a dark place. It is prohibited to apply sunlight or fluorescent light during the stor e.

17.3 OPERATION

(1) Do not connect, disconnect the module in the " Power On" condition.

(2) Power supply should always be turned on/off by following item 8.0 " Power on/off sequence ".

(3) Module has high frequency circuits. Sufficient suppression to the electromagnetic interference shall be done by system manufacturers. Grounding and shielding methods may be important to minimize the interference.

(4) The standard limited warranty is only applicable when the module is used for general notebook applications. If used for purposes other than as specified, BOE is not to be held reliable for the defective operations. It is strongly recommended to contact BOE to find out fitness for a particular purpose.

17.4 OTHERS

(1) Avoid condensation of water. It may result in improper operation or disconnection of electrode.

(2) Do not exceed the absolute maximum rating value. (the supply voltage variation, input voltage variation, Variation in part contents and environmental temperature, so on) Otherwise the module may be damaged.

(3) If the module displays the same pattern continuously for a long period of time, it can be the situation when The " image sticks" to the screen.

(4) This module has its circuitry PCB's on the rear or bottom side and should be handled carefully to avoid being stressed.

Appendix A

The Measurement Methods for the Dimensions of Module

Caliper:
a. Length of Outline
b. Width of Outline (Without/With PCB)
c. Thickness of Outline (Without/ With PCB)

Coordinate Measuring Machine:
CF Polarizer Size
Active Area Size
Active Area to Outline (Without Tape Wrinkle or Bulged)
Active Area to CF Polarizer
The Distance of Bracket Holes
P-Cover to Outline (Without Tape Wrinkle or Bulged)
Length of P-Cover
Connector Pin 1 to Outline (Without Tape Wrinkle or Bulged)

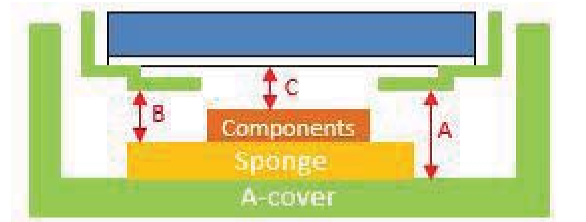
Height Gauge: The Different Height of Root and Top on the Bracket
(Need to Calculate From Bracket Angle Spec.)

Feeler Gauge: The Warpage Spec. of Module

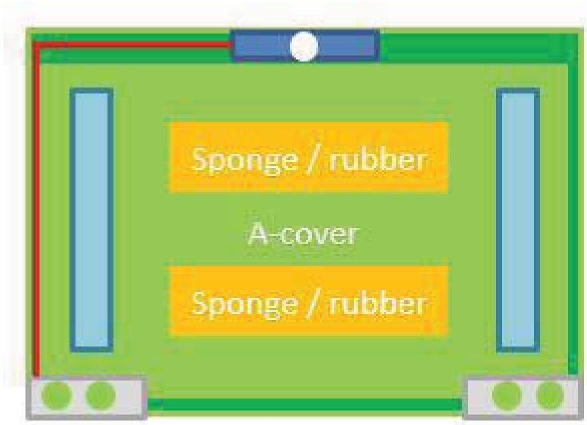
Notes:
Except the Critical Dimensions as Above, Other Dimensions are Measured by Coordinate Measuring Machine If Necessary.

Appendix B: System design Recommendation

LCM to A-Cover / sponges Z-gap



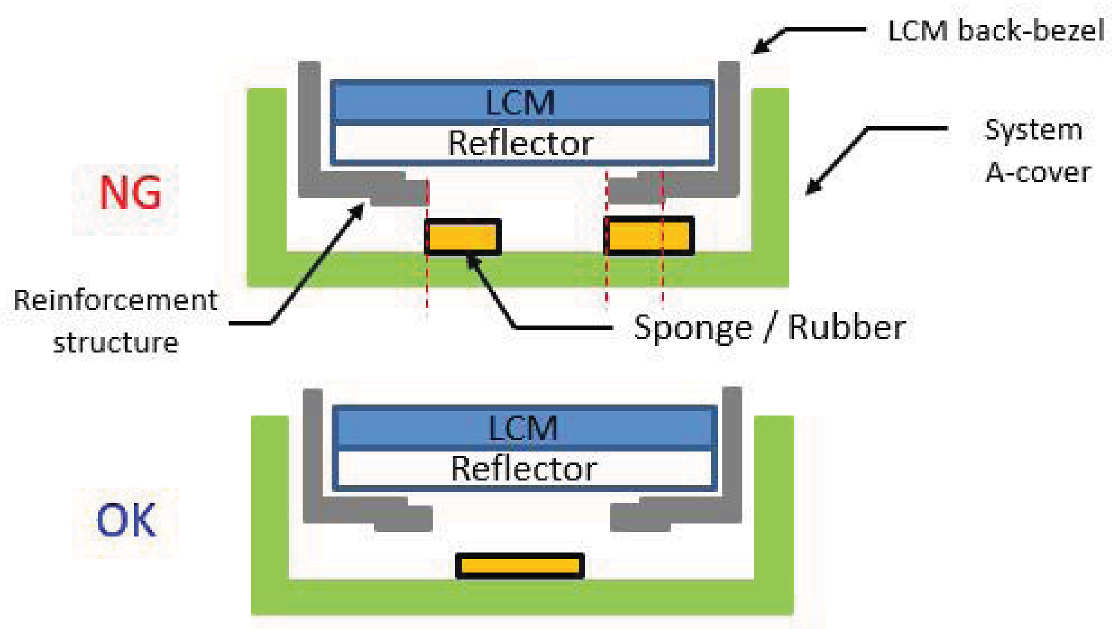
	Plastic Cover	Metal Cover
A	≥ 1.0mm	≥ 0.8mm
B	≥ 0mm	
C	> 0.5mm	



Purpose	The reflector area is very sensitive, BOE would suggest that design enough z-gap to decrease the risk of water ripple, white spots and other abnormal display

Appendix B: System design Recommendation

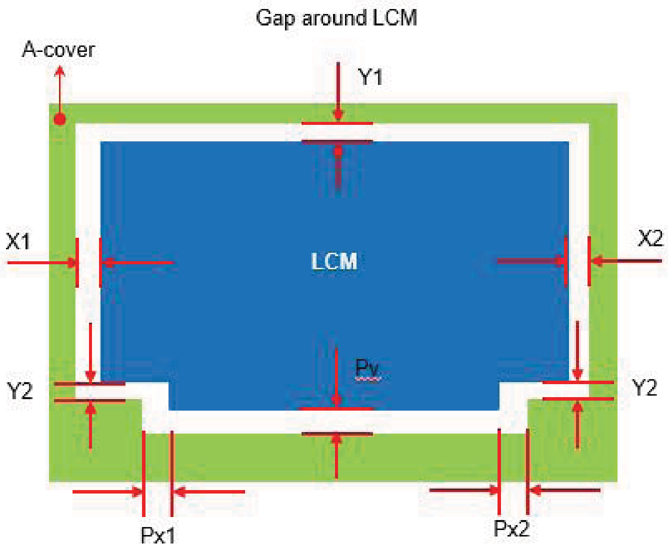
LCM to A-Cover / sponges z-gap



Purpose	If attach sponges or rubbers which correspond to white reflector area, it may cause white spot, pooling or other relative issues. BOE would suggest that attach wide range sponges / rubbers which can cover the LCM back-bezel opening

Appendix B: System design Recommendation

LCM to side wall / protrusions

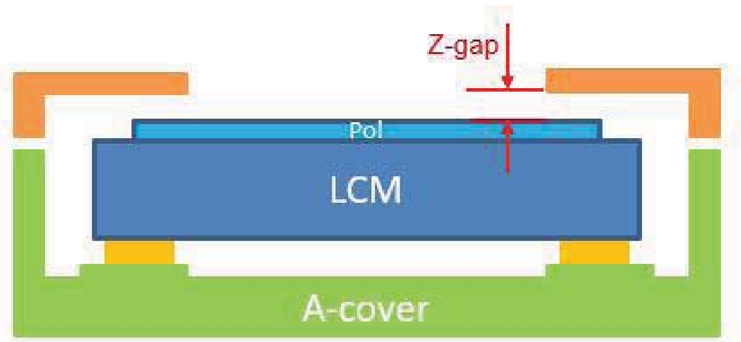


	Normal border (screws)	Narrow border (fix by tapes)
X1 / X2	Min: 0.45mm	Min: 0.35mm
Y1 / Y2	Min: 0.45mm	Min: 0.35mm
Px1 / Px2	Min: 0.55mm	
Py		

Purpose	BOE would suggest that design enough gap around LCM to prevent shock test failure, or interference, cell crack, abnormal display…etc. in the reliability test

Appendix B: System design Recommendation

LCM to B-cover z-gap

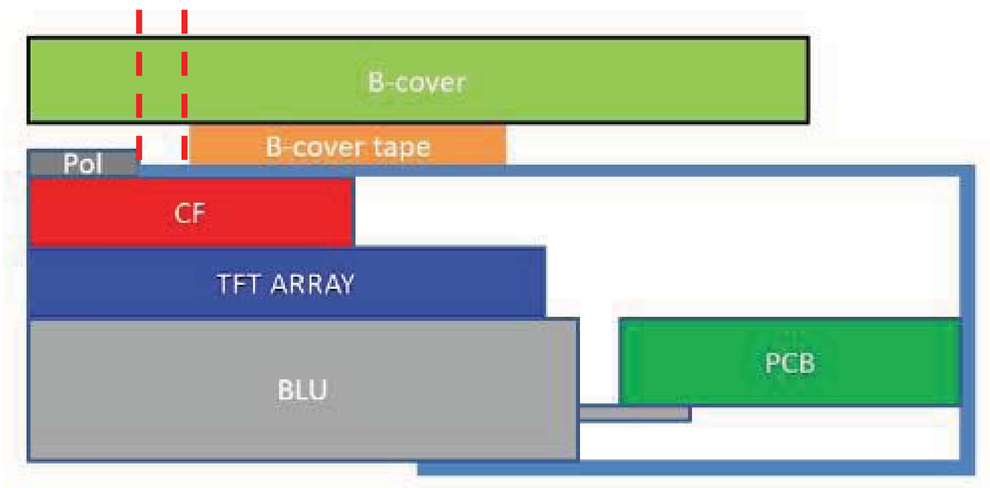


Bezel Tape	Z-Gap
Without	0.15 ~ 0.25mm
With	0.15 ~ 0.20mm

Purpose	Too less z-gap between system B-cover and LCM top pol has high risk that may cause cell crack, pooling, light leakage and other issues

Appendix B: System design Recommendation

B-cover tape to top pol edge

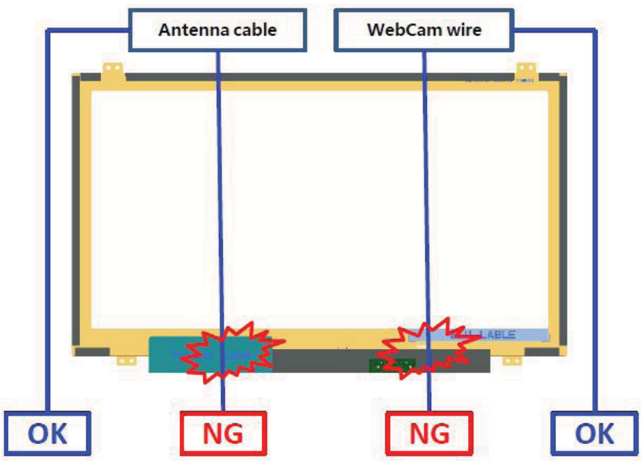


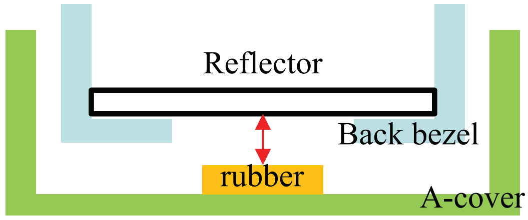
If attach b-cover and LCM with tapes,
Please let tapes to be located out of top pol edges 0.4mm away on 4 sides

Purpose	To avoid the B-cover tape override top pol then cause pooling or light leakage issue

Appendix B: System design Recommendation

Antenna Cable & Webcam wire

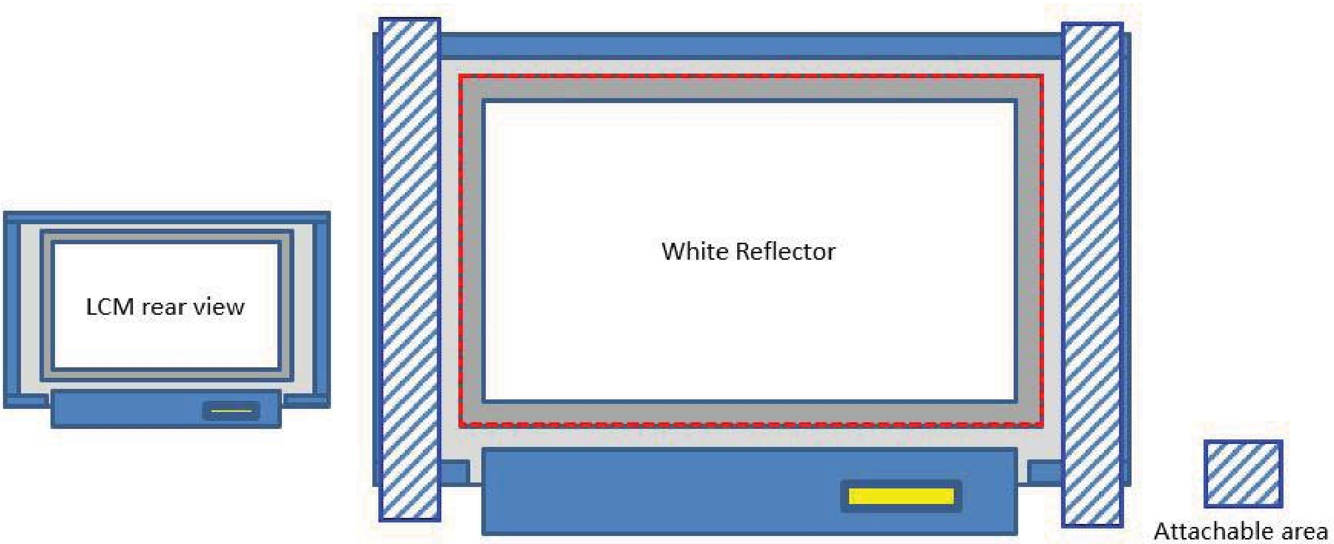




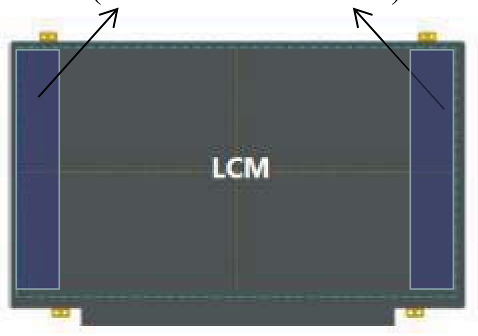
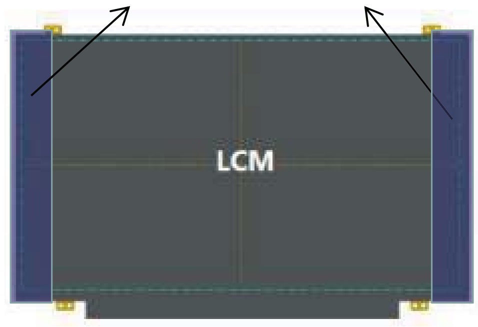
If sponge within the reflector area is necessary, we suggest that the gap between reflector and sponge is more than 0.5mm

Purpose	1. BOE would suggest that do not set Antenna or WebCam cable / wire go behind LCM to avoid backpack test, hinge test ,twist test or pogo test with abnormal display 2. If the cable / wire is necessary to go behind LCM, please make a groove with rounds or chamfers to protect the cable / wire, or attach with higher sponges / rubbers adjacent to the cable / wire route 3. Suggest that attach the cable / wire with tapes to A-cover 4. Do not attach anything with LCM reflector area. If attach cable / wire with LCM reflector area, it may cause pooling, white spot, light leakage and other related issues

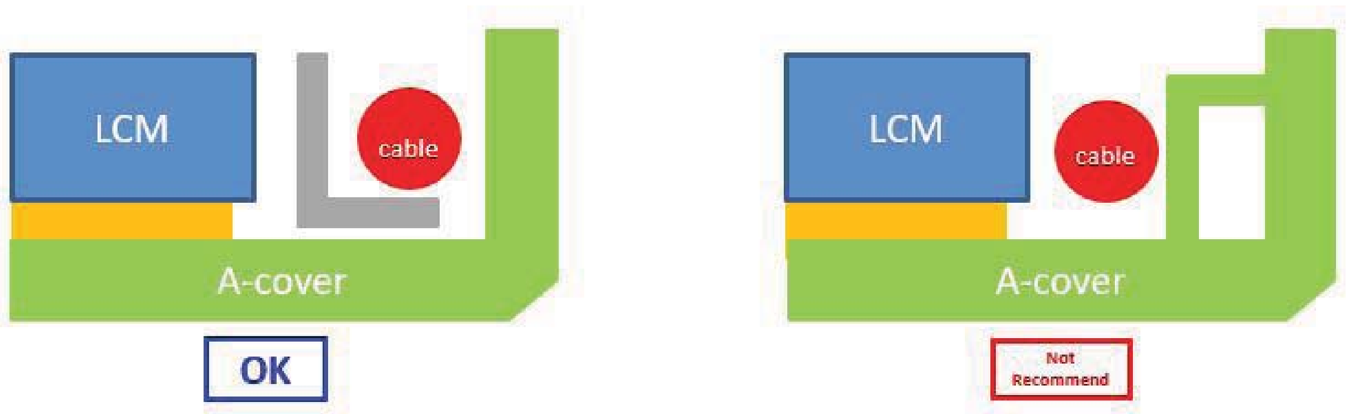
Appendix B: System design Recommendation

LCM paste area	
	
Purpose	If use the stretch remove tapes to fix LCM with A-cover, please set the stretch remove tapes correspond to the LCM back-bezel and do not let the tapes override the back-bezel's level step of opening

Appendix B: System design Recommendation

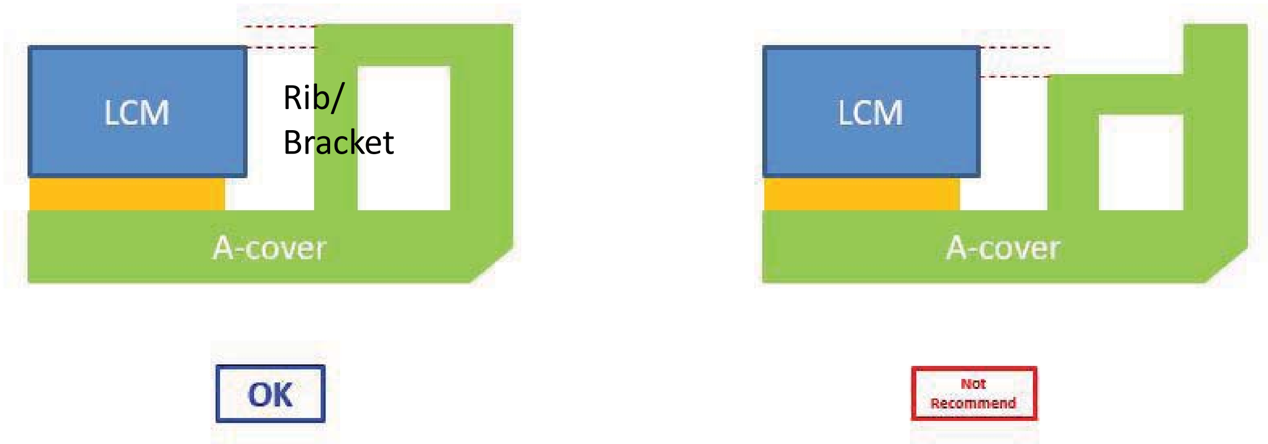
LCM pressable area	
 NG	 OK
Purpose	1. If LCM is fixed on A-cover by using the press jig during assembling. 2. To avoid panel broken the design of pressure head of press jig can not only pin on cell panel. The pressure head needs to pin on the LCM frame, which the LCM frame can share the pressure of the pressing head.

Appendix B: System design Recommendation

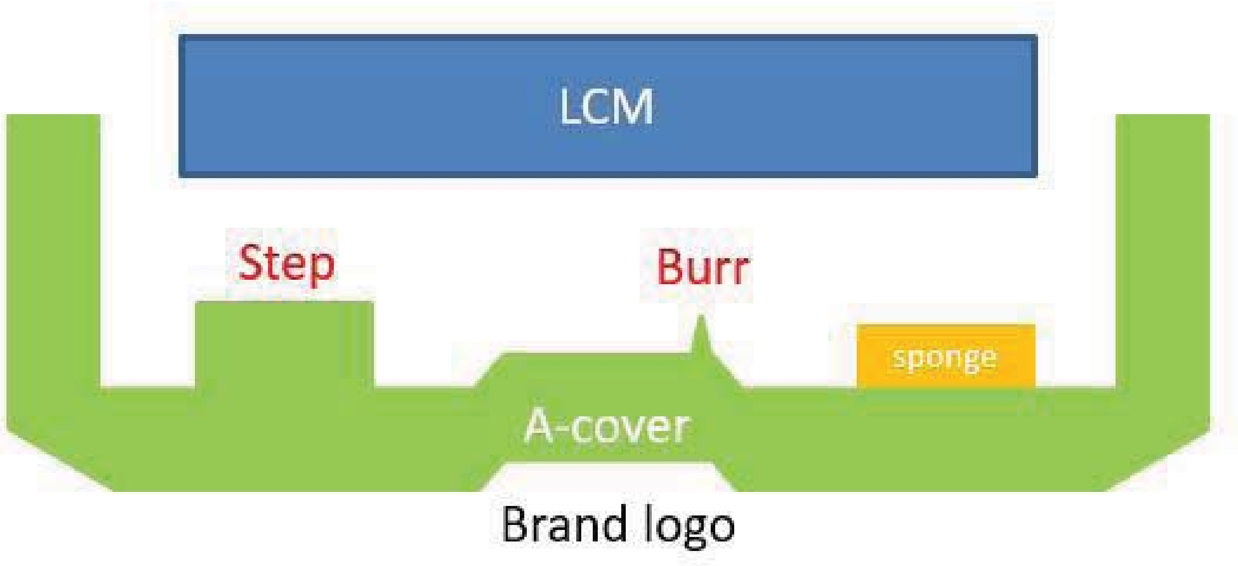
Wire setting	
	
Purpose	Wires should be placed between protrusions/side wall and A-cover. If place the wires between LCM and Protrusions/side wall, it may interfere with LCM when assembling, or even cause LCM broken in reliability test.

BOE	PRODUCT GROUP	REV	ISSUE DATE
	Customer Spec	Rev. P0	2024.02.19

Appendix B: System design Recommendation

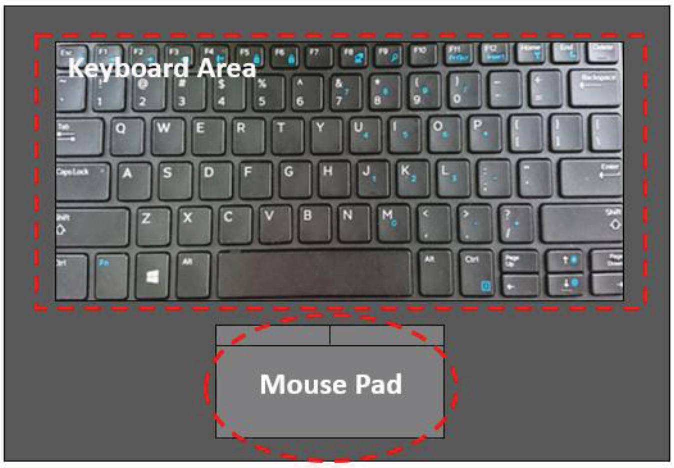
A-cover strength	
	
Purpose	1. BOE would recommend that structural Rib/Bracket height is higher than LCM, in order to avoiding pressures to LCM. 2. The L-shape Bracket is recommended.

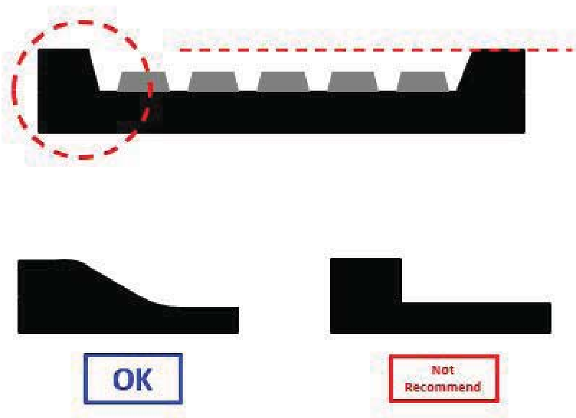
Appendix B: System design Recommendation

System A-cover Inner Surface	
	
Purpose	There should not exist any burr, segment gap or protrusions beside Logo, which may cause White Spot or Glass Broken by stress concentration.

Appendix B: System design Recommendation

Keyboard area & Mouse pad	
Purpose	The transition surface between keyboard and mouse pad should be smooth and without vertical steps\ too large level steps

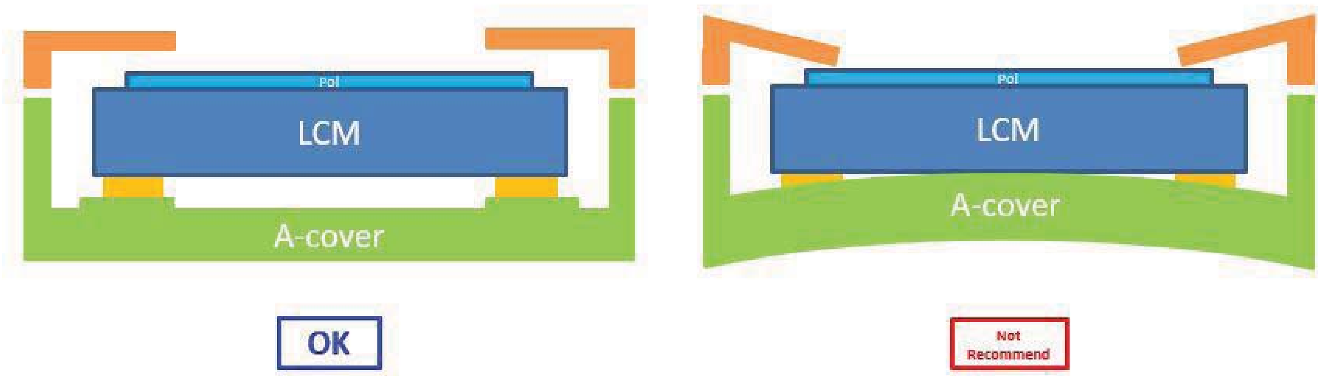




BOE	PRODUCT GROUP	REV	ISSUE DATE
	Customer Spec	Rev. P0	2024.02.19

Appendix B: System design Recommendation

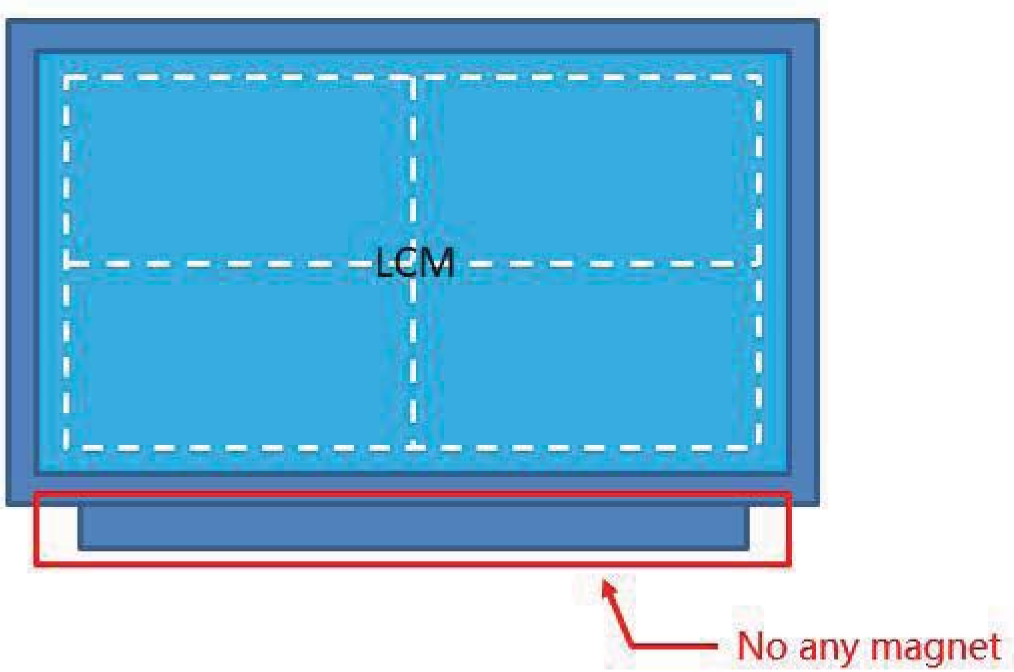
System cover reliability



Purpose	1. No interference between system and LCM in assembly process except compressible grounding gaskets 2. The permanent deformation which caused by Reliability test is not allowed to contact LCM

Appendix B: System design Recommendation

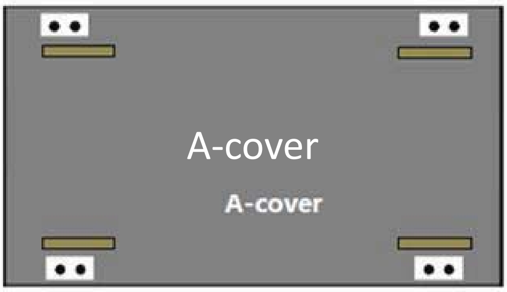
A/B-cover near LCD PCBA

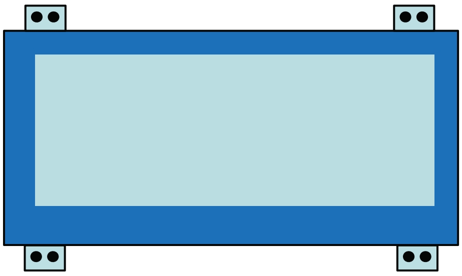


Purpose	There should not been any magnet object close to LCM PCBA, it may cause physical or electricity noise issue

Appendix B: System design Recommendation

A-cover add sponges on Boss side wall

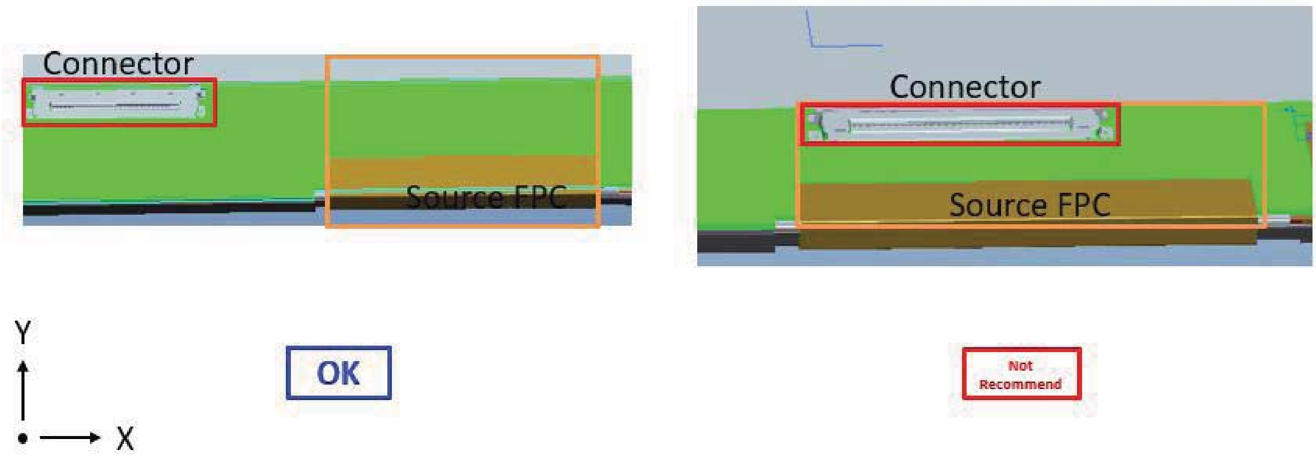




Purpose	BOE would suggest to attach Sponges to the side-wall of the Boss column of A-cover to reduce the risk of panel broken in assembling process.

Appendix B: System design Recommendation

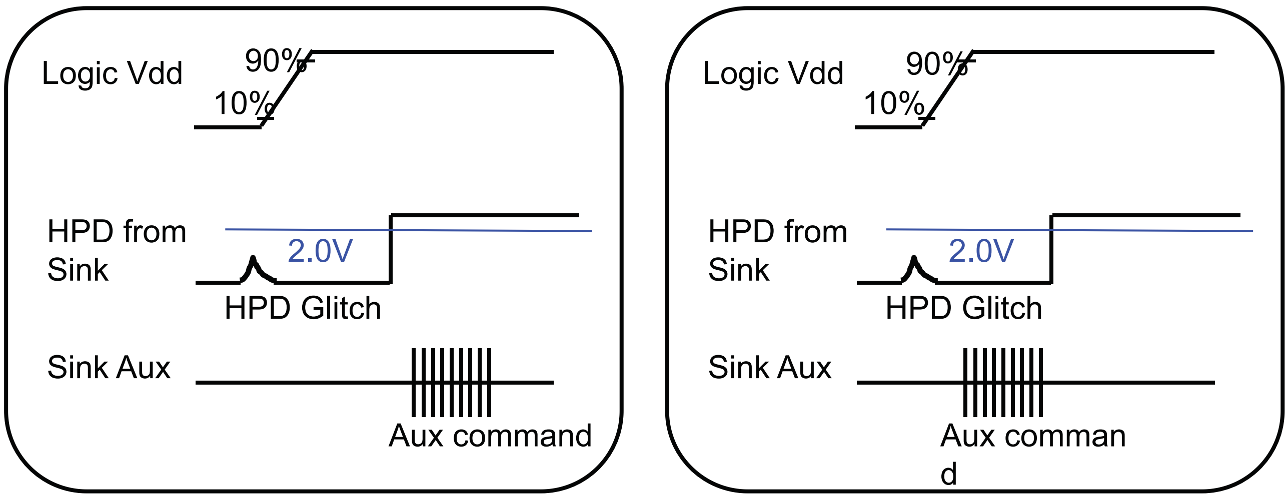
LCM to A-Cover / sponges z-gap



Purpose	Bent type product: The System Connector should not overlap with LCM FPC in X-direction, it may cause FPC lead broken during system connector plug and un-plug process (Panel FPC Bonding location is related to Mask and can not be changed easily)

Appendix C : System design Recommendation

HPD Signal recognition



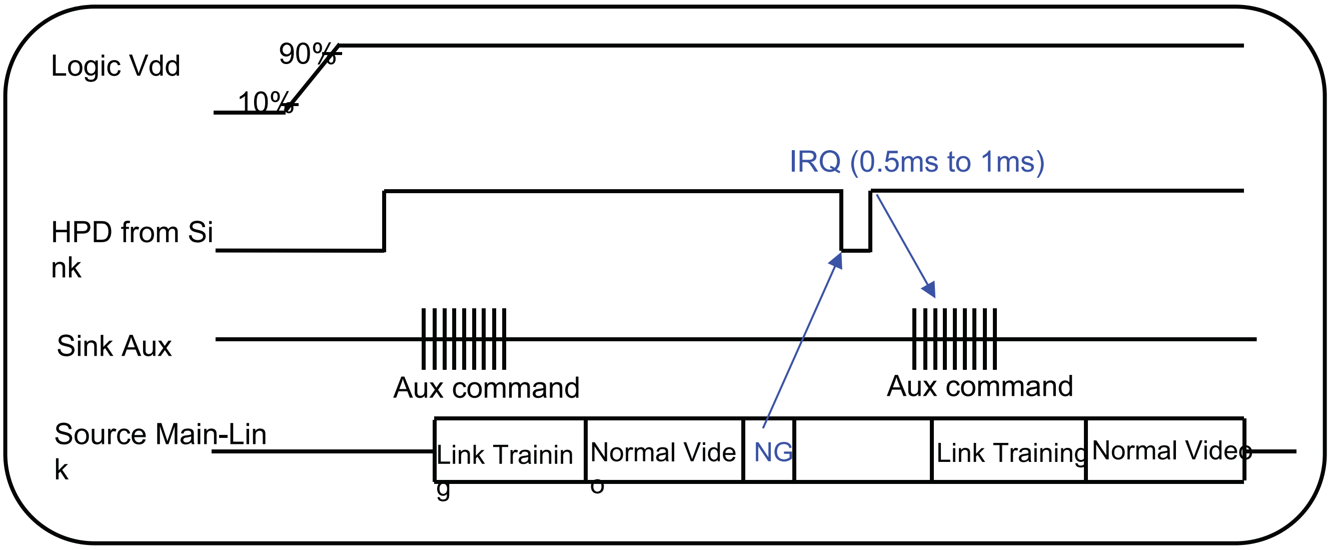
Normal Signal (Ignore HPD Glit ch)

Abnormal Signal

Purpose	When HPD glitch voltage less than 2.0(V), system signal can't output AUX command data.

Appendix C : System design Recommendation

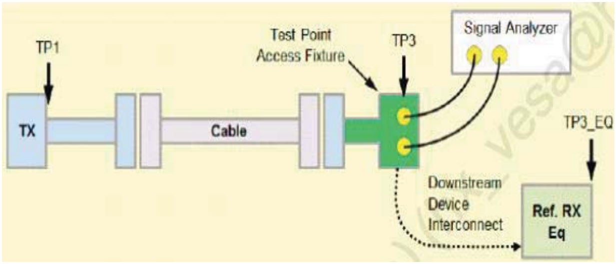
HPD Signal Definition IRQ (Interrupt Request)



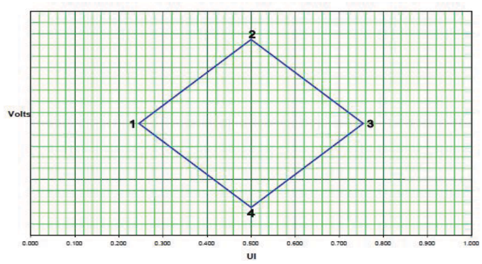
Purpose	When HPD signal low than 0.5ms to 1ms, the source device should check sink status field from the DPCD and take link training again.

Appendix C : System design Recommendation

Main link eye diagram of TP3



Measured TP3 on LCM connector.



Downstream Device Mask at TP3

	UI	Voltage
1	0.246	0
2	0.5	0.075
3	0.755	0
4	0.5	-0.075

Eye for TP3 at HBR

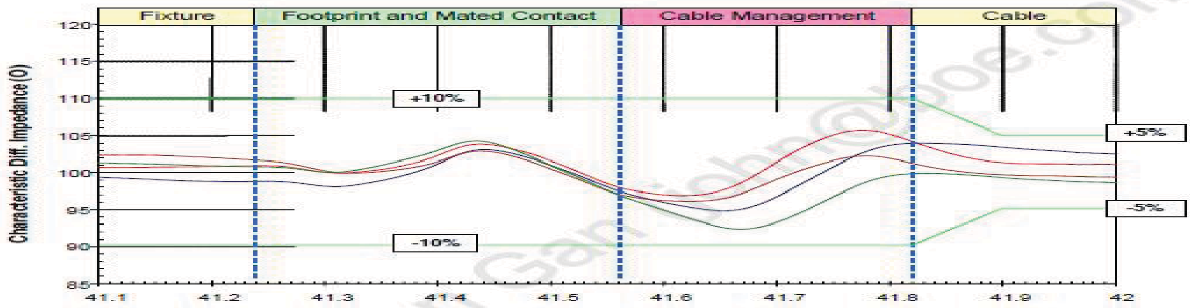
	UI	Voltage
1	0.375	0
2	0.5	0.023
3	0.625	0
4	0.5	-0.023

Eye for TP3 at RBR

Purpose	1. Main Link EYE Diagram should meet TP3 point of VESA. 2. The measure method is through access fixture.

Appendix C : System design Recommendation

Impedance Profile through a DP Connector



Differential Impedance Profile Measurement Data Example

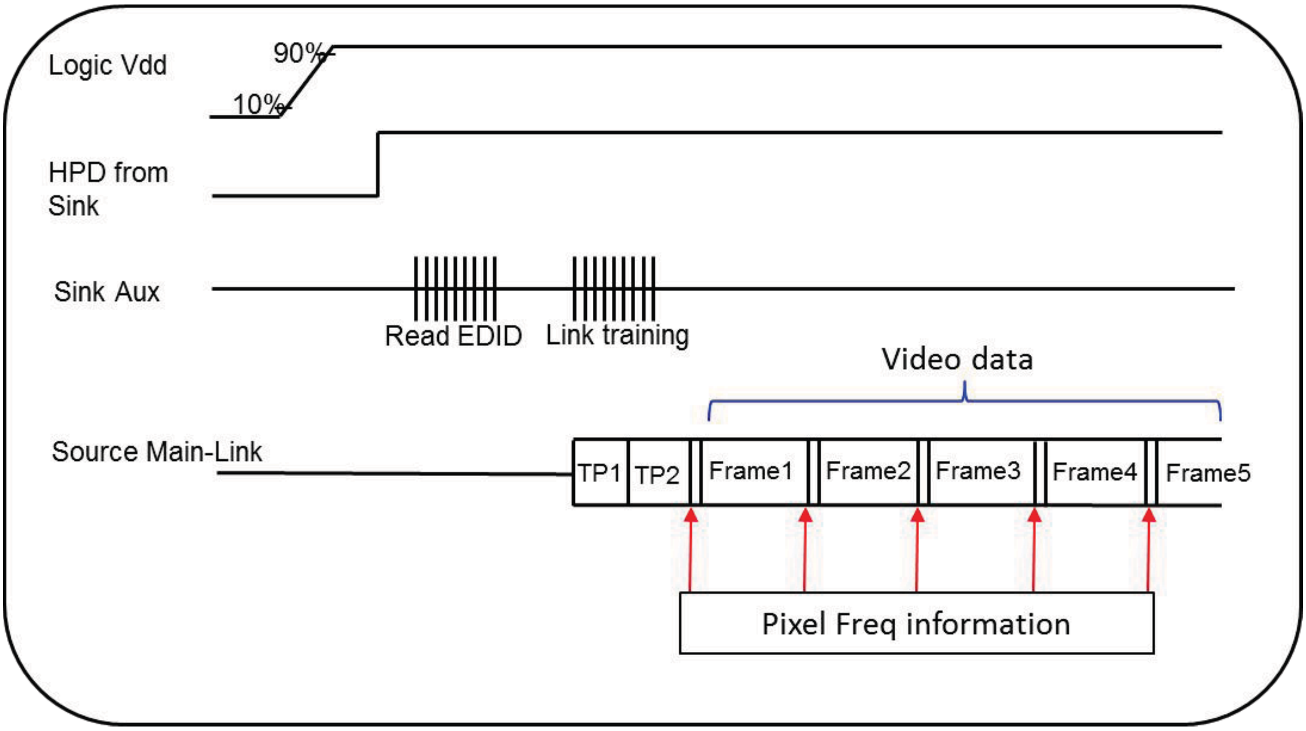
Segment	Differential Impedance Value	Maximum Tolerance
Fixture	100Ω/VESA	±10%
Connector	100Ω/VESA	±10%
Wire management	100Ω/VESA	±10%
Cable	100Ω/VESA	±5%

Impedance Profile Values for Cable Assembly

Purpose	Cable Impedance Profile 100ohm for Cable Assembly

Appendix C : System design Recommendation

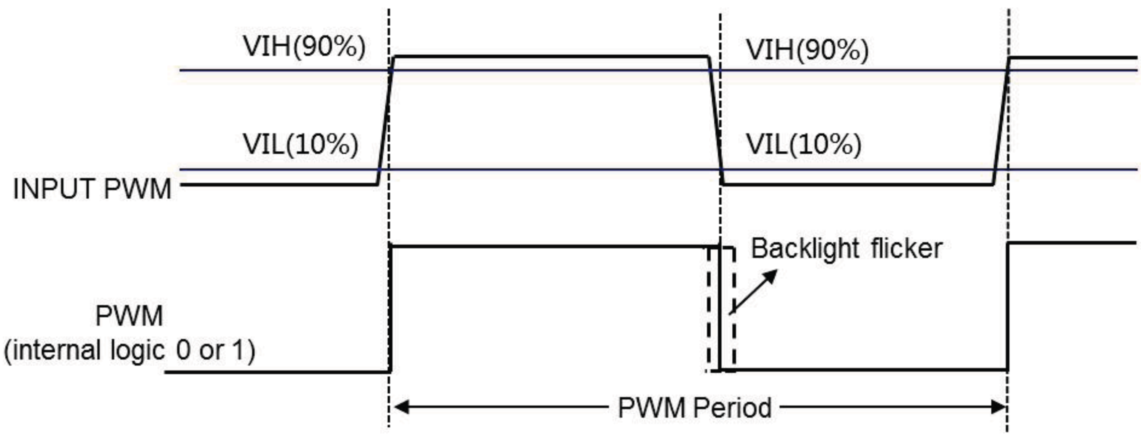
Main Link Pixel Freq information value of MSA data



Purpose	1. It need to fix pixel freq information value of MSA data output to prevent the initial abnormal pixel freq information value from incoming after power on. 2. BOE can read DPCD to check this value. Ex: BIOS is 1.62G , but into windows is 2.7G.

Appendix C : System design Recommendation

System Input PWM Rising/Falling time



Example:

Freq	Cycle Time	PWM Rising Time	PWM Falling Time
200Hz	5ms	≤1us	≤1us
1KHz	1ms	≤200ns	≤200ns

Purpose	1. LED driver need to calculate the duty cycle of input PWM signal. 2. To avoid backlight flicker visible on LCD, system input PWM suggest : PWM rising ≤ 200ppm*cycle time ; PWM falling ≤ 200ppm*cycle time.